Calibeating Prediction-Powered Inference

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Abstract

We study semisupervised mean estimation with a small labeled sample, a large unlabeled sample, and an available black-box prediction model that may be miscalibrated. A standard approach in this setting is augmented inverse-probability weighting (AIPW) (Robins et al., 1994), which is robust to prediction-model misspecification but may be inefficient when the prediction score is poorly calibrated. We introduce *Calibrated Prediction-Powered Inference*: post-hoc calibrate the prediction score on the labeled sample and then average the calibrated predictions over the pooled sample. This simple post-hoc step requires no retraining and can improve efficiency relative to using the original score directly. We study both linear and isotonic calibration. For isotonic calibration, we prove a first-order *calibeating-type* guarantee in both prediction and functional estimation: isotonic post-processing can improve on the original score and on simpler post-processing rules in predictive accuracy and estimator efficiency, and no further post-processing of the fitted isotonic score yields additional first-order improvement. For linear calibration, we show first-order equivalence to PPI++. We also show that the original PPI estimator is a special case of AIPW and can be inefficient when the prediction model is accurate, while PPI++ is AIPW with empirical efficiency maximization (Rubin and van der Laan, 2008). In simulations and real-data experiments, our calibrated estimators often outperform PPI and remain competitive with, or outperform, AIPW and PPI++. We provide an accompanying `Python` package, `ppi_aipw`.

1 Introduction

We study semisupervised mean estimation with one labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and one unlabeled dataset $\{\widetilde{X}_i\}_{i=1}^N$, where outcomes are observed only for a subset sampled completely at random. This is a classical missing-outcome problem: rich covariate information is available for many individuals, but the outcome of interest is observed only for a smaller subset. Such settings arise when gold-standard outcomes are expensive to collect but prediction scores are readily available, for example in randomized trials with prognostic models trained on historical data (Schuler et al., 2022), biobank analyses using phenotype predictors built from EHR or omics measurements (Hong et al., 2019; Zhang et al., 2019), and computational biology pipelines that supplement slow or costly experimental measurements with model-based predictions (Angelopoulos et al., 2023a; Zrnic and Candès, 2024a). The goal is to use the larger unlabeled sample to improve estimation of a population mean while accounting for prediction error. This setting has long been studied in missing-data and semiparametric statistics, where methods such as AIPW, TMLE, and related debiased estimators allow flexible machine-learning predictors to be used with valid inference (Robins et al., 1994; van der Laan and Robins, 2003; Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). More recently, related ideas have reappeared in the machine-learning literature under the name prediction-powered inference (Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024a). Closely related questions also arise in covariate adjustment for randomized trials, where machine-learning predictors trained on auxiliary data can

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improve precision in small samples (Hansen, 2008; Rubin and van der Laan, 2008; Moore and van der Laan, 2009a; Lin, 2013; Benkeser et al., 2021; Schuler et al., 2022; Demirel et al., 2024; Jin, 2026).

A standard approach is to use the labeled sample to debias a plug-in estimator based on a prediction model. If $m(X)$ is a black-box prediction score, the plug-in estimator averages $m(X)$ over the pooled covariate sample, while the labeled sample is used to estimate the residual correction term $\mathbb{E}\{Y - m(X)\}$. For example, the augmented inverse-probability weighted (AIPW) estimator (Robins et al., 1994) averages $m(X)$ over both labeled and unlabeled samples and then adds the average residual over the labeled sample, whereas the PPI estimator (Angelopoulos et al., 2023a) averages $m(X)$ only over the unlabeled covariates before applying the same residual correction. The latter is generally less efficient because it discards information in the labeled covariates. Thus, while these methods protect against bias from imperfect prediction models, they typically treat the prediction score itself as fixed. In practice, however, useful black-box scores are often systematically too high, too low, or too extreme. Such misalignment may arise because the model was trained on a different population, or because of temporal drift, distribution shift, or regularization bias.

This motivates post-hoc calibration of the prediction score. Rather than correcting only the marginal bias $\mathbb{E}\{Y - m(X)\}$, one can use the labeled sample to transform the score so that its numerical values better align with the outcomes they are meant to predict (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005). An ideal target is a calibrated score $m_n^\star(X)$ satisfying $\mathbb{E}[Y \mid m_n^\star(X)] = m_n^\star(X)$. Under this condition, $\mathbb{E}[m_n^\star(X)] = \mathbb{E}[Y]$, so the corresponding plug-in estimator is automatically unbiased for the population mean. This is attractive in practice because a useful prediction model may already be available, while a relatively small labeled sample can often calibrate its output without retraining the underlying model (Roth, 2022). Classical post-processing methods include Platt scaling (Platt et al., 1999), linear calibration (Mincer and Zarnowitz, 1969), histogram binning (Zadrozny and Elkan, 2001), and isotonic regression (Zadrozny and Elkan, 2002). Such calibration is especially relevant for modern machine-learning predictors, such as neural networks, which are often informative yet systematically miscalibrated (Bella et al., 2010; Guo et al., 2017; Wang, 2023).

An early grounded toy example makes the mechanism concrete. We take a public LLM-evaluation benchmark with human correctness labels, focus on its MATH slice, and ask how accurately one can estimate the average human correctness from a small labeled sample plus many unlabeled evaluator scores. The raw evaluator margin has real signal but is poorly scaled for the human labels, so using it directly in PPI yields only modest gains. A one-dimensional recalibration fit on the labeled pairs restores the probability scale, sharply reduces held-out squared error, and yields materially shorter confidence intervals for the mean. In this sense, calibration improves label efficiency: achieving the same uncertainty with the raw score or with labeled-only estimation would require more human labels. Figure 1 visualizes this mechanism in the resulting one-mean toy.

Our proposal is therefore simple: fit a black-box regression model, calibrate it on the labeled sample, and then average the calibrated predictions over the pooled covariate sample. This remains a simple "predict, calibrate, and average" procedure, but we show that it also admits an exact debiased interpretation. In particular, the resulting estimator is exactly equal to an AIPW estimator based on the calibrated score, with the residual correction encoded by the calibration step itself. Because calibration can improve the score, it can also improve efficiency relative to AIPW based on the uncalibrated model. Calibration therefore converts a simple plug-in rule into a debiased estimator with attractive efficiency properties.

Contributions. We make three main contributions.

First, in Section 3, we study *calibrated prediction-powered inference* for semisupervised mean estimation with a small labeled sample and a larger unlabeled one. This approach accommodates familiar post-hoc calibration methods, including linear calibration, Platt scaling, histogram binning, and isotonic regression; see, in particular, Sections 3.2 and 3.3. We focus in particular on two practical estimators, linearly calibrated PPI and isotonic-calibrated PPI, which are simple to implement, tuning-parameter-free, and easy to add to an existing prediction pipeline without retraining the original model. Isotonic calibration is especially appealing because it learns the mean-squared-error-optimal monotone transformation of the prediction score.

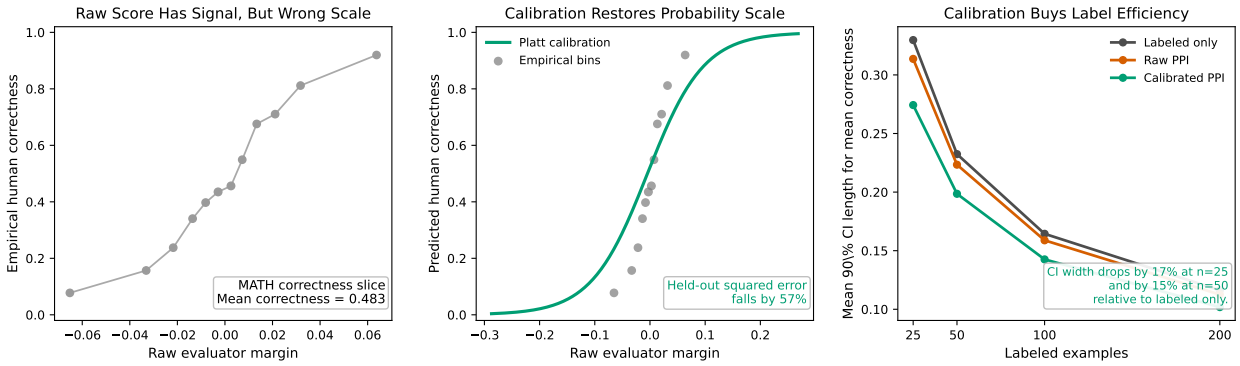

Figure 1: Grounded one-mean toy derived from the MATH slice of a public LLM-evaluation benchmark with human correctness labels, using ArmoRM as the automatic evaluator score. Left: the raw evaluator margin has clear signal about correctness, but it is not on the right numerical scale. Middle: a Platt map fit on 100 labeled examples restores the probability scale and lowers held-out squared error by about 58%. Right: in Monte Carlo with $N = 2000$ unlabeled scores, calibrated PPI yields about 17% shorter 90% Wald intervals than labeled-only estimation at $n = 25$ and about 15% shorter intervals at $n = 50$, while raw PPI provides only modest gains.

Second, we argue that calibration can provide three important statistical benefits. First, it can improve the prediction score and thereby increase the precision of point estimates and confidence intervals. Second, it automatically debiases the calibrated plug-in estimator, because that estimator is exactly equal to the AIPW estimator constructed from the calibrated score. Third, calibration can produce a fitted score that admits no further first-order improvement by post-processing: within the relevant class of post-processing maps, no additional transformation can improve predictive performance or reduce estimator variance to first order. We view this as a form of *calibeating* in the spirit of Foster and Hart (2023). We distinguish between two forms. *Prediction calibeating* means that the calibrated score is post-processing-optimal for prediction to first order within the relevant class. *Functional-estimation calibeating* means that the resulting estimator is post-processing-optimal for asymptotic efficiency to first order within that class.

Third, we develop asymptotic theory for linearly calibrated and isotonic-calibrated PPI estimators. For linear calibration, we show that PPI++ is asymptotically equivalent both to linear calibration and to linear regression adjustment using $m(X)$ as a prognostic score. This clarifies the connection between PPI++ and classical covariate-adjustment methods in randomized trials and missing-at-random settings. For isotonic-calibrated PPI, Section 4 establishes asymptotic normality, valid inference, and efficiency results. In particular, we show that isotonic-calibrated PPI achieves the smallest asymptotic variance among estimators based on monotone transformations of the original score, and is therefore at least as efficient as AIPW and PPI using the uncalibrated score. We also show that no further post-processing of the fitted score can improve predictive performance at first order, and that the resulting estimator is first-order equivalent to the oracle estimator based on the conditional mean of the outcome given the calibrated score.

Our calibrated PPI framework can be viewed as a semisupervised specialization of the calibrated debiased machine learning framework developed by van der Laan et al. (2024c). Our estimators can also be interpreted as targeted minimum loss estimators (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011), with calibration playing the role of the targeting step.

Section 2 reviews the two-sample semisupervised model, the AIPW, PPI, and PPI++ estimators, and the main efficiency concepts used throughout the paper. Section 5 presents synthetic and real-data experiments.

Code. The accompanying `Python` package, reproduction code, and documentation are publicly available at github.com/Larsvanderlaan/ppi-aipw. An interactive package website with examples and API documentation is available at larsvanderlaan.github.io/ppi-aipw/. Self-contained code is provided in Appendix M.

1.1 Related work

Although PPI has been presented as a recent development, the underlying statistical problem and several closely related estimators fall within a much broader literature on semiparametric debiasing and regression adjustment, including work on missing data, causal inference, and debiased machine learning. In particular, the use of machine-learning predictions in missing-at-random settings to improve efficiency is closely connected to the rich literature on covariate adjustment in randomized trials and causal inference (Moore and van der Laan, 2009a; VanderWeele et al., 2019; Benkeser et al., 2021; Li et al., 2023; Van Lancker et al., 2024; Højbjerre-Frandsen et al., 2025), while "power-tuning" ideas can be viewed as a form of empirical efficiency maximization (Rubin and van der Laan, 2008).

Semiparametric statistics and missing data. This paper builds on foundational ideas from the missing-data and semiparametric literatures. Rubin (1976) formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes. Semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures (Levit, 1975; Pfanzagl and Wefelmeyer, 1985; Hasminskii and Ibragimov, 1979; Klaassen, 1987; Bickel et al., 1993). In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003; Tsiatis, 2006), often with doubly robust structure (Bang and Robins, 2005). As a result, the efficiency theory for missing-data models under missingness or coarsening at random is now well understood, and the class of regular asymptotically linear estimators in these models can be characterized through their influence functions (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003; Tsiatis, 2006).

Debiased machine learning and targeted learning. A large body of work combines the semiparametric framework above with machine-learning-based nuisance estimation, often under the label of debiased machine learning. Flexible learning methods within estimating equations and AIPW-style procedures already appear in van der Laan and Robins (2003). Targeted minimum loss estimation (TMLE), or targeted learning, updates an initial nuisance estimator in a parameter-targeted way so that the resulting plug-in estimator is debiased; in the present setting, the main nuisance is the outcome regression (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Gruber and Van Der Laan, 2009; Hoffman, 2020; Ross et al., 2025). These approaches are often combined with sample splitting and cross-fitting to accommodate generic machine-learning estimators (Schick, 1986; Klaassen, 1987; Zheng and Van Der Laan, 2010; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). Double machine learning emphasizes similar ingredients—orthogonality, cross-fitting, and influence-function expansions—for inference with modern nuisance estimators (Chernozhukov et al., 2018a, 2022a), and can be understood within the same semiparametric framework (van der Laan, 2019; Díaz, 2020; Chen et al., 2026). For sufficiently stable estimators, cross-fitting can itself be relaxed (Chen et al., 2022). Closely related work studies balancing-weight estimators (Hainmueller, 2012; Imai and Ratkovic, 2014; Zubizarreta, 2015; Lendle et al., 2015; Chattopadhyay et al., 2020; Hejazi and van der Laan, 2023), which often admit equivalent regression-adjustment or augmented representations (Chattopadhyay and Zubizarreta, 2023; Bruns-Smith et al., 2025; Rotnitzky et al., 2025). In completely missing-at-random settings, including semisupervised learning and randomized trials, these approaches allow flexible black-box nuisance estimation while still supporting valid inference under appropriate conditions (Moore and van der Laan, 2009a,b; Rosenblum and van der Laan, 2009; Højbjerre-Frandsen et al., 2025; Højbjerre-Frandsen and Schuler, 2026).

Semisupervised inference and improving power in randomized trials. Related work studies semisupervised or surrogate-assisted estimation in settings where the primary outcome is observed only on a limited subset, while auxiliary outcomes or surrogate measurements are available more broadly (Pepe, 1992; Pepe et al., 1994; Chen, 2000; Chen et al., 2008; Cheng et al., 2021; Ji et al., 2025; Kallus and Mao, 2025). Two-sample semisupervised inference is a special case of two-stage sampling designs (Scott and Holt, 1982; Rose and van der Laan, 2011; Hejazi et al., 2021; Qiu et al., 2026) and, more generally, of data fusion, for which a substantial semiparametric efficiency literature has been developed (Li and Luedtke, 2023; Li et al., 2025; Graham et al., 2024; Xu et al., 2025). More recent papers have used the term *prediction-powered inference* (PPI) for the semisupervised mean-estimation setting in which a black-box predictor is combined with a small labeled sample and a large unlabeled sample (Angelopoulos et al., 2023a,b; Zrnic and Candès,

2024a; Xu et al., 2025; Song et al., 2026; Poulet et al., 2025; Wang et al., 2025; Lee and Kim, 2026). These papers also helped popularize such methods for a broader machine-learning audience and provided accessible software. Closely related questions have long been studied in randomized trials, where prognostic scores or other regression adjustments learned from larger historical, external-control, or auxiliary data sources are used to improve precision in smaller trials (Hansen, 2008; Rosenblum and van der Laan, 2009; Moore and van der Laan, 2009a,b; Moore et al., 2011; VanderWeele et al., 2019; Benkeser et al., 2021; Burger et al., 2021; Schuler et al., 2022; Li et al., 2023; Balzer et al., 2024; Van Lancker et al., 2024; Højbjerre-Frandsen et al., 2025). Methodologically, however, PPI estimators are often best viewed as instances or refinements of existing semiparametric procedures. For example, PPI++ is an AIPW estimator (Robins et al., 1994; van der Laan and Robins, 2003) obtained by empirical efficiency maximization (Rubin and van der Laan, 2008) over a scaling parameter, while cross-PPI (Zrnic and Candès, 2024a) is a cross-fitted AIPW estimator (Zheng and Van Der Laan, 2010; Chernozhukov et al., 2018a). Likewise, semiparametric efficient PPI (Ji et al., 2025) draws heavily on semiparametric efficiency results developed in Robins et al. (1994) and van der Laan and Robins (2003), specialized to the semisupervised setting. More fundamentally, the ability to combine black-box prediction models with valid inference follows from the double robustness of AIPW when the missingness probability is known, and is therefore not unique to semisupervised mean estimation (Bang and Robins, 2005; Seaman and Vansteelandt, 2018; Rotnitzky et al., 2021).

Calibrated DML and Recalibrated PPI. In our mean-estimation setting, the closest precursor is the calibrated debiased machine-learning framework of van der Laan et al. (2024c), which studies calibration of the outcome regression in doubly robust inference. Our setting can be viewed as a semisupervised specialization of that perspective, adapted to prediction-powered inference and geared toward improving efficiency in randomized designs. This connection also motivates our term *calibrated PPI*. A closely related paper is Ji et al. (2025), who study recalibrated prediction-powered inference for general estimating equations. Their method targets an oracle correction to the imputed loss, rather than applying a calibration map directly to the predictions. In the mean-estimation setting under squared loss, this oracle correction reduces to the conditional mean of the outcome given the initial prediction score, which yields the efficient estimator in the reduced model based only on that score. This conditional mean can be viewed as an oracle calibrator, so the two approaches are closely related. The difference lies in the estimators, emphasis, and theory. Ji et al. (2025) use flexible machine learning to estimate this oracle quantity and thereby pursue semiparametric efficiency (Robins et al., 1994). They do not study specific calibration procedures, and their estimator remains an AIPW estimator. By contrast, we study simple, commonly used post-hoc calibration maps applied directly to a fixed score, and ask what can be guaranteed even without consistent estimation of the oracle target. Our estimators retain a simple plug-in form, yielding lightweight methods that can be added to an existing prediction pipeline without retraining. We also develop calibration-specific results: an exact AIPW interpretation of the plug-in estimator, first-order equivalence between linear calibration and PPI++, and a first-order calibeating-type guarantee for isotonic calibration.

From missing data to causal inference. From the perspective of estimation and efficiency theory, semisupervised mean estimation with missing-at-random outcomes is equivalent to randomized experiments, controlled trials, and survey-sampling designs once the missingness indicator is reinterpreted as a treatment or sampling indicator, so that the relevant potential outcomes are viewed as missing (Rubin, 1978; van der Laan and Robins, 2003; Ding and Li, 2018; Mozer, 2026). This equivalence allows tools from covariate adjustment, causal inference, and model-assisted estimation to be transferred directly to the semisupervised setting; see, for example, Mozer (2026) for results relating prediction-powered inference to model-assisted survey-sampling estimators. The two-sample formulation we study is also equivalent to a one-sample i.i.d. model (Li and Luedtke, 2023). In particular, our methods apply to inference on the counterfactual mean $\mathbb{E}[Y(1)]$ in a randomized trial with binary treatment, where treated units form the labeled sample and control units contribute covariate information only; see Appendix D.

Calibration. A final relevant thread concerns calibration. Classical post-processing methods such as Platt scaling (Platt et al., 1999; Cox, 1958; Gupta and Ramdas, 2023), linear calibration (Mincer and Zarnowitz, 1969; Chernozhukov et al., 2018b; Leng and Dimmery, 2021), isotonic regression (Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005), and histogram binning (Zadrozny and Elkan, 2001; Gupta and Ramdas, 2021) improve predictive reliability without retraining the underlying model. The idea has deeper roots in

the forecasting literature (Mincer and Zarnowitz, 1969; Lichtenstein et al., 1977; Vovk, 1992; Vovk et al., 2005; Gneiting et al., 2007; Lambert, 2011); see also the historical discussion in Lee et al. (2023). More recent work studies calibration under broader losses and prediction tasks (Jung et al., 2021; Noarov and Roth, 2023; van der Laan et al., 2023; Whitehouse et al., 2024; van der Laan and Alaa, 2025), as well as applications to conformal prediction (van der Laan and Alaa, 2024, 2025), importance-weight stabilization (Gutman et al., 2022; Deshpande and Kuleshov, 2023; Ballinari and Bearth, 2024; van der Laan et al., 2024b), treatment effect estimation (Chernozhukov et al., 2018b; van der Laan et al., 2023; Whitehouse et al., 2024), and value prediction in offline reinforcement learning (van der Laan and Kallus, 2025; van der Laan et al., 2025b). Distribution-free guarantees for histogram binning are developed by Gupta and Ramdas (2021), while van der Laan et al. (2023); van der Laan and Alaa (2025) establish asymptotic distribution-free guarantees for isotonic calibration.

2 Two-Sample Semisupervised Setup

2.1 Data structure and notation

We observe two independent datasets,

$$\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n \qquad \text{and} \qquad \mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N.$$

The labeled sample consists of i.i.d. draws from a distribution P_0 on $\mathcal{X} \times \mathbb{R}$, while the unlabeled sample consists of i.i.d. draws from the distribution $P_{0,X}$ of X under P_0. Our target is the population mean

$$\psi_0 := \mathbb{E}_{P_0}[Y], \qquad \mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x].$$

This can also be written as

$$\psi_0 = \int y \, dP_0(x, y) = \int \mu_0(x) \, dP_{0,X}(x).$$

For functions $a : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ and $b : \mathcal{X} \to \mathbb{R}$, define the empirical means over the labeled and unlabeled samples by

$$\mathbb{P}_n^L a := \frac{1}{n} \sum_{i=1}^n a(X_i, Y_i), \qquad \mathbb{P}_N^U b := \frac{1}{N} \sum_{j=1}^N b(\widetilde{X}_j).$$

For readability, we will sometimes write $\mathbb{P}_n^L\{a(X, Y)\}$ instead of $\mathbb{P}_n^L a$. When a function depends only on X, we also write

$$\mathbb{P}_n^L\{b(X)\} := \frac{1}{n} \sum_{i=1}^n b(X_i).$$

Throughout, let $M := n + N$ and $\rho_n := n/M$ denote the labeled fraction. Then

$$\frac{1}{n+N} \left\{ \sum_{i=1}^n f(X_i) + \sum_{j=1}^N f(\widetilde{X}_j) \right\} = \rho_n \mathbb{P}_n^L f + (1 - \rho_n) \mathbb{P}_N^U f.$$

In other words, the empirical mean of $f(X)$ over the pooled covariate sample is a weighted average of the empirical means from the labeled and unlabeled samples.

Assumption 1 (Two-sample design)**.** The following hold:

(i) $\mathcal{D}_L$ is i.i.d. from P_0, and $\mathcal{D}_U$ is i.i.d. from $P_{0,X}$.

(ii) $\mathcal{D}_L$ and $\mathcal{D}_U$ are independent.

(iii) Y has finite second moment under P_0.

(iv) With $M := n + N$, the labeled fraction $\rho_n = n/M$ satisfies $\rho_n \to \rho_0$ for some $\rho_0 \in (0, 1)$.

Remark 1. The labeled sample carries the outcome information, while the unlabeled sample sharpens estimation of the marginal covariate distribution. The pooled i.i.d. missing-data formulation in Appendix K is a special case of this two-sample setup.

2.2 Review of AIPW, PPI, and semiparametric efficiency

The AIPW class. We begin with the augmented inverse-probability weighted (AIPW) estimator in the unrestricted two-sample model (Robins et al., 1994). It is helpful to view AIPW not as a single estimator, but as a family of estimators indexed by a score function $f : \mathcal{X} \to \mathbb{R}$ with finite second moment. For any such f, define

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{1}$$

This estimator combines a plug-in term that imputes Y from $f(X)$ with a residual correction from the labeled sample. For any fixed f, $\widehat{\psi}(f)$ is unbiased in finite samples for $\psi_0 = \mathbb{E}_{P_0}[Y]$. In particular, $f(X) = 0$ yields the empirical mean estimator $\mathbb{P}_n^L\{Y\}$. Thus, the choice of f affects efficiency, through the estimator's variance, but not validity, in the sense that the resulting confidence intervals remain valid. This makes AIPW a "safe" estimator (Xu et al., 2025). This property can be viewed as a particularly simple instance of double robustness[1] (Bang and Robins, 2005). The same basic principle underlies the use of flexible machine learning for covariate adjustment in randomized trials while retaining valid inference (Moore and van der Laan, 2009a,b; Schuler et al., 2022; Højbjerre-Frandsen et al., 2025).

Asymptotic linearity and efficiency. To characterize the leading large-sample error in the two-sample model, we decompose the estimation error into one contribution from the labeled sample and one from the unlabeled sample. We say that $\widehat{\psi}$ is asymptotically linear with influence pair (D^L, D^U) if

$$\widehat{\psi} - \psi_0 = \frac{1}{M}\sum_{i=1}^{n} D^L(X_i, Y_i) + \frac{1}{M}\sum_{j=1}^{N} D^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

where $\mathbb{E}_{P_0}[D^L(X, Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D^U(\widetilde{X})] = 0$. By the central limit theorem and the independence of the two samples,

$$\sqrt{M}\,(\widehat{\psi} - \psi_0) \rightsquigarrow N\Big(0,\ \rho_0 \mathrm{Var}_{P_0}\{D^L(X, Y)\} + (1 - \rho_0)\mathrm{Var}_{P_{0,X}}\{D^U(\widetilde{X})\}\Big).$$

The influence pair therefore determines the estimator's asymptotic variance and provides a convenient way to compare efficiency across different choices of f. The next result uses this representation to characterize the AIPW class, identify the efficient influence pair in the unrestricted two-sample model, and thereby obtain the semiparametrically efficient variance bound (Bickel et al., 1993).

Proposition 1 (Influence pair for the AIPW class)**.** *Under Assumption 1, for any fixed $f : \mathcal{X} \to \mathbb{R}$ with finite second moment, the estimator $\widehat{\psi}(f)$ is asymptotically linear with influence pair (D_f^L, D_f^U), where*

$$D_f^L(X, Y) = \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\},$$
$$D_f^U(\widetilde{X}) = \tilde{f}(\widetilde{X}) - \psi_0,$$

with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$ denoting the recentered version of $f(X)$. The resulting asymptotic variance of $\sqrt{M}\,(\widehat{\psi}(f) - \psi_0)$ is

$$\mathrm{Var}\{\mu_0(X)\} + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] + \frac{1 - \rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

[1] In more general missing-at-random settings with unknown, covariate-dependent missingness probabilities, the analogous AIPW estimator remains doubly robust asymptotically, but it no longer enjoys the same finite-sample unbiasedness as in the present known-ρ_n setting (Seaman and Vansteelandt, 2018).

This variance is minimized over the AIPW class when $\tilde{f} = \mu_0$, in particular at $f = \mu_0$. Therefore, the efficient influence pair for $\psi_0 = \mathbb{E}_{P_0}[Y]$ in the unrestricted two-sample model is $(D^L_{\mu_0}, D^U_{\mu_0})$, and the corresponding asymptotic variance is the efficiency bound[2].

A key implication of Proposition 1 is that the asymptotic variance of the AIPW estimator based on f exceeds the efficiency bound by $\frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X)-\mu_0(X))^2\big]$. Thus, efficiency within the AIPW class is governed entirely by how well $f(X) - \mathbb{E}_{P_0}[f(X)]$ approximates $\mu_0(X) - \mathbb{E}_{P_0}[Y]$ in mean squared error, with the optimum attained at $f = \mu_0$. In turn, full efficiency generally requires the full covariate information X, or at least a summary rich enough to recover $\mu_0(X)$ (Robins et al., 1994; Robins and Rotnitzky, 1995). This motivates using a flexible black-box prediction score $m(X)$ for regression adjustment in practice.

AIPW, PPI, and empirical efficiency maximization. We now record several familiar choices of f within the class (1). The most common choice is $f(X) = m(X)$, where $m(X)$ is a prediction score, yielding the usual augmented inverse-probability weighted (AIPW) estimator (Robins et al., 1994):

$$\widehat{\psi}_{\text{AIPW}} = \rho_n \mathbb{P}^L_n\{m(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m(\widetilde{X})\} + \mathbb{P}^L_n\{Y - m(X)\}.$$

Equivalently,

$$\widehat{\psi}_{\text{AIPW}} = \frac{1}{M}\left\{\sum_{i=1}^n m(X_i) + \sum_{j=1}^N m(\widetilde{X}_j) + \frac{1}{\rho_n}\sum_{i=1}^n \{Y_i - m(X_i)\}\right\},$$

where ρ_n^{-1} is the inverse probability of observing the outcome.

Another estimator in this class was considered by Angelopoulos et al. (2023a) under the name PPI,[3] and we adopt that terminology here. It corresponds to the score choice

$$f(X) = \frac{m(X)}{1-\rho_n}.$$

Substituting this choice into (1) gives

$$\widehat{\psi}_{\text{PPI}} = \mathbb{P}^U_N\{m(\widetilde{X})\} + \mathbb{P}^L_n\{Y - m(X)\}.$$

Relative to $\widehat{\psi}_{\text{AIPW}}$, $\widehat{\psi}_{\text{PPI}}$ uses only the unlabeled sample in the plug-in term and rescales the score by the factor $(1-\rho_n)^{-1} > 1$. By Proposition 1, this helps only if the rescaled score is closer to $\mu_0(X)$ in mean squared error. Thus, PPI is inefficient when $m = \mu_0$, and is typically less efficient when m is already reasonably accurate. When ρ_n is small and m is sufficiently inaccurate, however, neither estimator is necessarily more efficient than the other (Han, 2012; Rotnitzky et al., 2012).

Because both $\widehat{\psi}_{\text{AIPW}}$ and $\widehat{\psi}_{\text{PPI}}$ use scores in the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, it is natural to choose the scaling factor data-adaptively rather than fixing it in advance. One way to do so is to minimize the estimator's empirical influence-function variance over this class:

$$\sigma_n^2(f) := \rho_n \mathbb{P}^L_n\left[\left\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\right\}^2\right] + (1-\rho_n)\mathbb{P}^U_N\left[\left\{f(\widetilde{X}) - \widehat{\psi}(f)\right\}^2\right].$$

That is, one chooses λ to minimize $\sigma_n^2(f)$ over $f(X) = \lambda m(X)$, a procedure known as empirical efficiency maximization (Rubin and van der Laan, 2008; van der Laan and Robins, 2003). The resulting estimator, called PPI++ by Angelopoulos et al. (2023b), is an AIPW estimator with an empirically rescaled prediction rule $\hat{\lambda} m(X)$, where $\hat{\lambda}$ estimates the population-optimal scaling coefficient

$$\lambda^\star = \frac{\operatorname{Cov}\big(Y, m(X)\big)}{\operatorname{Var}\big(m(X)\big)}. \tag{2}$$

[2]Appendix A shows that every regular estimator is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators and the semiparametric efficiency bound in this model.

[3]The PPI and PPI++ estimators are algebraically equivalent to the model-assisted survey-design estimators of Cassel et al. (1976) and Särndal et al. (2003), respectively; see Mozer (2026) for a review.

The same principle extends beyond simple rescaling. For example, one can minimize $\sigma_n^2(f)$ over richer classes such as $\{\lambda m(X)+\beta_{\text{lin}}^\top X : \lambda \in \mathbb{R},\ \beta_{\text{lin}} \in \mathbb{R}^d\}$, over monotone transformations of the score, or over a reproducing kernel Hilbert space (Appendix C). More generally, the same criterion can be used in cross-validation to choose among candidate estimator classes (Rubin and van der Laan, 2008). In particular, if many prediction models $\{m_1(X), m_2(X), \dots\}$ are available, one could learn a weighted linear combination by minimizing the empirical variance.

3 Calibrated Prediction-Powered Inference

Section 2.2 showed that AIPW provides a safe debiasing template, while the choice of score $f(X)$ determines efficiency. We now introduce the *calibrated prediction-powered inference* framework, which improves an available prediction score $m(X)$ through post-processing rather than retraining. Using the labeled sample, we learn a transformation of $m(X)$ within a chosen class and then use the resulting calibrated score to construct an estimator of the target mean. As we will show, this estimator admits both a simple plug-in form and an exact AIPW representation. Different choices of calibration class lead to different calibrated PPI estimators, including the linear and isotonic procedures studied below.

3.1 General estimator and exact AIPW representation

Let $m : \mathcal{X} \to \mathbb{R}$ denote a black-box regression model. A calibration procedure uses the labeled scores $m(X_1), \dots, m(X_n)$ and outcomes $Y_1, \dots, Y_n$ to learn a post-hoc map f_n, and then defines the calibrated regression model

$$m_n^\star := f_n \circ m.$$

The resulting predictions on the labeled and unlabeled samples are $m_n^\star(X_i)$ for $i = 1, \dots, n$ and $m_n^\star(\widetilde{X}_j)$ for $j = 1, \dots, N$. Intuitively, the map f_n updates the original score $m(X)$ so that its numerical values better align with the outcome.

This post-processing step has two main goals. First, it aims to improve the original prediction score m through a simple transformation, without retraining the underlying model, ideally by improving calibration without sacrificing predictive performance. Second, it is designed so that the resulting plug-in estimator is automatically unbiased, without requiring an explicit residual correction.

Given the post-processed score $m_n^\star$, define the calibrated PPI estimator by

$$\widehat{\psi}_{\text{cal}} := \frac{1}{n+N}\left\{\sum_{i=1}^{n} m_n^\star(X_i) + \sum_{j=1}^{N} m_n^\star(\widetilde{X}_j)\right\}. \tag{3}$$

Equivalently,

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\}.$$

To distinguish the class used to learn the post-processing map from the class used to assess the resulting prediction score, we introduce two function classes. Let $\mathcal{H}$ be an optimization class of real-valued maps $h : \mathbb{R} \to \mathbb{R}$, and let $\mathcal{F}$ be a linear calibration class of real-valued maps $f : \mathbb{R} \to \mathbb{R}$ containing the constant functions and the identity map. We learn the post-processing map f_n by empirical risk minimization over $\mathcal{H}$, that is,

$$f_n \in \arg\min_{h \in \mathcal{H}} \mathbb{P}_n^L\big[(Y - h(m(X)))^2\big]. \tag{4}$$

We then ask what calibration property the resulting post-processed score $m_n^\star := f_n \circ m$ satisfies with respect to the class $\mathcal{F}$.

We say that $m_n^\star$ is *empirically calibrated over* $\mathcal{F}$ if, for every $f \in \mathcal{F}$,

$$\mathbb{P}_n^L\big[(Y - m_n^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - f(m_n^\star(X)))^2\big], \tag{5}$$

where $\mathbb{P}_n^L$ denotes the empirical mean over the labeled sample. In words, once $m_n^\star$ has been formed, no further transformation of $m_n^\star$ within $\mathcal{F}$ can improve the empirical squared-error fit.

Because $\mathcal{F}$ is linear, this condition is equivalent to the first-order optimality conditions

$$\mathbb{P}_n^L\big[f(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0, \qquad f \in \mathcal{F}. \tag{6}$$

Since $\mathcal{F}$ contains the constant functions and the identity map, these conditions imply, in particular,

$$\mathbb{P}_n^L\{Y - m_n^\star(X)\} = 0, \qquad \mathbb{P}_n^L\big[m_n^\star(X)\{Y - m_n^\star(X)\}\big] = 0.$$

The first identity is an empirical mean-calibration condition: the residuals of the calibrated score average to zero on the labeled sample, yielding the exact AIPW representation below. The second shows that $m_n^\star$ is already optimal over the one-dimensional scaling class $\{\lambda m_n^\star : \lambda \in \mathbb{R}\}$, so no rescaling of $m_n^\star$ can further reduce empirical squared error.

The next result gives the basic structural property of calibrated PPI estimators.

Theorem 1 (Exact AIPW representation under empirical calibration)**.** *Suppose the calibrated predictions satisfy equation (6), where $\mathcal{F}$ contains all affine maps. Then, for any $\widehat{w} \in \mathcal{F}$,*

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big]. \tag{7}$$

In particular, taking $\widehat{w} \equiv 1$ yields the exact AIPW representation

$$\widehat{\psi}_{\text{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m_n^\star(X)\}.$$

If, in addition, $\widehat{w}_n^\star \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\, f(m_n^\star(X))\big] = \rho_n \mathbb{P}_n^L\{f(m_n^\star(X))\} + (1-\rho_n)\mathbb{P}_N^U\{f(m_n^\star(\widetilde{X}))\}, \qquad f \in \mathcal{F}, \tag{8}$$

then

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))Y\big]. \tag{9}$$

Theorem 1 shows that empirical calibration does more than improve prediction: it gives the calibrated PPI estimator an exact debiased representation. Specifically, $\widehat{\psi}_{\text{cal}}$ is exactly an AIPW estimator with score $m_n^\star$. The same calibration conditions also yield an equivalent balancing-weight representation, under which $\widehat{\psi}_{\text{cal}}$ is a weighted average of Y over the labeled sample, with the unlabeled sample entering only through the weights. This representation makes clear how unlabeled information is incorporated and enables familiar weighting diagnostics, such as balance checks and inspection of the weight distribution.

Full versus linear calibration. We focus on two main cases. In the nonparametric case, $\mathcal{F}$ is the class of all transformations $f : \mathbb{R} \to \mathbb{R}$, which we call *full calibration*. At the population level, a predictor is fully calibrated if no further transformation can improve its mean squared prediction error. In practice, however, the fitted post-processing map f_n is obtained by minimizing empirical squared error over a chosen optimization class $\mathcal{H}$, which need not coincide with $\mathcal{F}$. The class $\mathcal{H}$ acts as a regularizer: without such a restriction, one could overfit the labeled sample, achieving perfect empirical fit but poor prediction and poor population-level calibration. For example, $\mathcal{H}$ may consist of piecewise constant or monotone functions, as in histogram regression or isotonic regression (Section 3.2). In the parametric case, both $\mathcal{H}$ and $\mathcal{F}$ are restricted to affine transformations $t \mapsto a + bt$, which we call *linear calibration*. This corresponds to fitting the working linear model $a + b\,m(X)$ for Y (Section 3.3).

Why calibration can improve efficiency. The representation in Theorem 1 explains why calibration yields a valid debiased estimator; the remaining question is why it may also improve precision. To see this, suppose that the fitted post-processing map f_n converges to a population minimizer $g_0 \in \mathcal{H}$, and hence that the post-processed score $m_n^\star = f_n \circ m$ converges to

$$m_0(X) = g_0\{m(X)\}, \qquad g_0 \in \arg\min_{g \in \mathcal{H}} \mathbb{E}_{P_0}\big[\{Y - g(m(X))\}^2\big].$$

Algorithm 1 Isotonic-calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $\hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i - f(T_i)\}^2$
3: Set $m^\star_{n,\text{iso}}(X_i) = \hat{f}(T_i)$ and $m^\star_{n,\text{iso}}(\widetilde{X}_j) = \hat{f}(\widetilde{T}_j)$
4: Form $\widehat{\psi}_{\text{iso}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}$
6: Set $\widehat{\text{SE}}^2_{\text{iso}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{iso}}$

Thus, m_0 is the best predictor obtainable from the original score m using post-processing maps in $\mathcal{H}$. This squared-error criterion is also the relevant first-order efficiency criterion: minimizing empirical mean-squared error over $\mathcal{H}$ is asymptotically equivalent to empirical efficiency maximization for the resulting calibrated PPI estimator; see Appendix C. When $m_n^\star$ is close to m_0, $\widehat{\psi}_{\text{cal}}$ therefore has the same first-order behavior as the AIPW estimator using m_0. Proposition 1 then implies that its asymptotic variance is no larger than that of the AIPW estimator based on any $\theta \circ m$ with $\theta \in \mathcal{H}$. In particular, if $\mathcal{H}$ contains the identity map, as it does for affine or monotone transformations, then m itself is among the admissible post-processings. If $\mathcal{H}$ contains the oracle map $t \mapsto \mathbb{E}_{P_0}[Y \mid m(X) = t]$, then $m_0(X) = \mathbb{E}_{P_0}[Y \mid m(X)]$ and the estimator is semiparametric efficent in the reduced model based on the score $m(X)$ (Ji et al., 2025). It follows that calibration can improve efficiency insofar as m_0 provides a better regression adjustment than m.

3.2 Isotonic-calibrated PPI

We now describe two simple second-stage calibration methods that fit naturally into the $(\mathcal{H}, \mathcal{F})$ framework: histogram binning and isotonic regression (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005). Both are fit using the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$, but they use different optimization classes $\mathcal{H}$.

For histogram binning, $\mathcal{H}$ is the class of piecewise-constant maps on a prespecified partition of the score space (Gupta et al., 2020; Gupta and Ramdas, 2021). One may take $\mathcal{F} = \mathcal{H}$. In fact, because the fitted score is constant within each bin, any further transformation of that score is also piecewise constant on the same partition. Thus histogram binning is empirically calibrated not only over $\mathcal{H}$, but also fully calibrated. In this sense, a post-processing map learned over a restricted class $\mathcal{H}$ can still yield full empirical calibration.

For isotonic calibration, $\mathcal{H} = \mathcal{F}_{\text{iso}}$, where $\mathcal{F}_{\text{iso}}$ denotes the class of nondecreasing real-valued functions. The calibrator is fit by isotonic regression via the pooled-adjacent-violators algorithm (Barlow and Brunk, 1972). The fitted score is piecewise constant on a data-adaptive partition of the score space determined by the pooled blocks. As with histogram binning, any further transformation remains constant on the same blocks, so isotonic calibration also yields full empirical calibration. We focus on isotonic regression because it does not require prespecified bins: the partition is learned adaptively from the data, and the method is lightweight, tuning-free, and easy to implement with widely available software.

The isotonic-calibrated PPI estimator is

$$\widehat{\psi}_{\text{iso}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\right\},$$

where

$$m^\star_{n,\text{iso}}(X) = \hat{f}\{m(X)\}, \qquad \hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i - f(m(X_i))\big\}^2.$$

The key calibration property of isotonic regression, and likewise of histogram binning, is the following.

Proposition 2 (Isotonic calibration is fully calibrated)**.** *For every function $h : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{P}_n^L\big[h(m^\star_{n,\mathrm{iso}}(X))\{Y - m^\star_{n,\mathrm{iso}}(X)\}\big] = 0. \tag{10}$$

Moreover,

$$\mathbb{P}_n^L\big[(Y - m^\star_{n,\mathrm{iso}}(X))^2\big] \le \mathbb{P}_n^L\big[(Y - m(X))^2\big].$$

Consequently, the representations in Theorem 1 hold with $\mathcal{F}$ taken to be the class of all real-valued functions.

Remark 2 (Implementation)**.** Isotonic regression is widely available in standard software. A standard algorithm is the pooled-adjacent-violators algorithm (PAVA) (Barlow and Brunk, 1972). Although it is generally fast, it may become slow at ultra-large sample sizes. A scalable alternative is to use regression tree software with monotonicity constraints, such as XGBoost or LightGBM (see Appendix M). An advantage of tree-based software is that the fit can be regularized by constraining or tuning the number of leaves, the maximum tree depth, or the minimum number of observations per leaf.

3.3 Linearly calibrated PPI

For small labeled samples, one may prefer smoother parametric calibration rules to histogram binning or isotonic regression. We focus on linear calibration, in which $\mathcal{H} = \mathcal{F} = \mathcal{F}_{\mathrm{lin}}$, where $\mathcal{F}_{\mathrm{lin}}$ denotes the class of affine maps $t \mapsto at + b$. Thus, the post-processing step is learned over the same class with respect to which calibration is assessed. Compared with isotonic regression, this imposes a more restrictive parametric form, akin in spirit to Platt scaling and temperature scaling (Platt et al., 1999; Guo et al., 2017), but it may be more stable when the labeled sample is small. Its guarantee is correspondingly narrower: it need not correct nonlinear miscalibration, but it is optimal within the class of affine transformations.

Let

$$\big(\hat{a}_{\mathrm{lin}}, \hat{b}_{\mathrm{lin}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2, \qquad m^\star_{n,\mathrm{lin}}(X) := \hat{a}_{\mathrm{lin}}\, m(X) + \hat{b}_{\mathrm{lin}}. \tag{11}$$

Equivalently, $m^\star_{n,\mathrm{lin}}$ is obtained by empirical risk minimization over the affine class $\mathcal{F}_{\mathrm{lin}}$. The associated linearly calibrated PPI estimator[4]is

$$\widehat{\psi}_{\mathrm{lin}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\right\}. \tag{12}$$

Because $\mathcal{F}_{\mathrm{lin}}$ contains the constant function and the identity map, the least-squares fit enforces empirical mean calibration and empirical orthogonality between the calibrated score and its residuals. Consequently, $\widehat{\psi}_{\mathrm{lin}}$ satisfies the exact AIPW representation and the no-further-calibration property within the affine class.

More generally, one can include additional covariates in the linear regression adjustment to further improve precision, provided there is sufficient labeled data (Algorithm 3).

Proposition 3 (Linear calibration as AIPW)**.** *The normal equations for equation (11) imply*

$$\mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{lin}}(X)\} = 0 \qquad \textit{and} \qquad \mathbb{P}_n^L\big[m^\star_{n,\mathrm{lin}}(X)\{Y - m^\star_{n,\mathrm{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^\star_{n,\mathrm{lin}}(X)$,

$$\widehat{\psi}_{\mathrm{lin}} = \rho_n \mathbb{P}_n^L\{m^\star_{n,\mathrm{lin}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{lin}}(\widetilde{X})\} + \mathbb{P}_n^L\big[\hat{w}(X)\{Y - m^\star_{n,\mathrm{lin}}(X)\}\big]. \tag{13}$$

The next result connects linear calibration to PPI++, a closely related estimator that rescales the original score by empirical efficiency maximization. In the mean-estimation problem, the two approaches are first-order equivalent, though they need not coincide exactly in finite samples. Intuitively, both target the same

[4]Appendix A shows that this estimator can equivalently be implemented by balancing on $(1, m(X))$ and then forming the corresponding weighted outcome estimator.

Algorithm 2 Linearly calibrated PPI

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{b}) \in \arg\min_{a,b} \sum_{i=1}^n \{Y_i - (aT_i + b)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a}T_i + \hat{b}$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \hat{b}$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

Algorithm 3 Linearly calibrated PPI with covariate adjustment

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{\beta}, \hat{b}) \in \arg\min_{a,\beta,b} \sum_{i=1}^n \{Y_i - (a + X_i^\top \beta + bT_i)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a} + X_i^\top \hat{\beta} + \hat{b}T_i$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a} + \widetilde{X}_j^\top \hat{\beta} + \hat{b}\widetilde{T}_j$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

one-dimensional adjustment, namely the optimal rescaling of $m(X)$, given by the population slope λ_0 in (2) from the linear regression of Y on $m(X)$. Let $\widehat{\psi}_{++}(\widehat{\lambda}_{++})$ denote the PPI++ estimator of Angelopoulos et al. (2023b), that is, the AIPW estimator based on the score $\widehat{\lambda}_{++}m(X)$, where $\widehat{\lambda}_{++}$ is obtained by empirical efficiency maximization over $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ (Rubin and van der Laan, 2008).

Proposition 4 (First-order equivalence of PPI++ and linear calibration)**.** *Assume Assumption 1 and that* $\mathrm{Var}\{m(X)\} > 0$ *and* $\mathbb{E}_{P_0}[m(X)^2] < \infty$*. Then*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}}\big)\left\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\right\}.$$

Moreover, $\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}} = O_p(n^{-1/2})$*, and*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = O_p\left(\frac{1-\rho_n}{\sqrt{\rho_n}}\,(n+N)^{-1}\right) = O_p\big((n+N)^{-1}\big).$$

In particular,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = o_p\big(M^{-1/2}\big).$$

Remark 3 (Relation to prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through covariate adjustment in small-sample settings (Hansen, 2008; Rosenblum and van der Laan, 2009; Lin, 2013; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Proposition 4 shows that PPI++ has the same first-order interpretation. Linear calibration can also be viewed as a special case of TMLE (Højbjerre-Frandsen and Schuler, 2026).

Remark 4 (Platt scaling adjustment)**.** The linear-calibration idea extends beyond squared-error loss. One can instead use logistic loss, Poisson loss, or, more generally, losses from canonical-link generalized linear models (Rosenblum and van der Laan, 2009). In particular, when $Y \in \{0,1\}$ and $m(X) \in (0,1)$ is an initial

predictor, one may apply Platt scaling (Cox, 1958; Platt et al., 1999) by fitting a logistic regression of Y on the logit of $m(X)$:

$$\big(\hat{a}_{\text{platt}}, \hat{b}_{\text{platt}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n} \Big[-Y_i \log \sigma\big(a \operatorname{logit}(m(X_i)) + b\big) - (1-Y_i)\log\Big(1 - \sigma\big(a \operatorname{logit}(m(X_i)) + b\big)\Big)\Big], \quad (14)$$

with calibrated predictor

$$m^{\star}_{n,\text{platt}}(X) := \sigma\big(\hat{a}_{\text{platt}} \operatorname{logit}(m(X)) + \hat{b}_{\text{platt}}\big),$$

where $\sigma(t) := (1+e^{-t})^{-1}$. Platt scaling may be preferred over linear regression calibration when predicted probabilities are systematically over- or under-confident, in which case the calibration curve is often approximately sigmoidal. More generally, outcomes can be rescaled to $[0,1]$ before applying this procedure. In online settings, calibration properties of Platt scaling are developed in Gupta and Ramdas (2023).

3.4 Cross-Fitting and Cross-Calibration

In the main exposition, we assume for simplicity that the initial black-box predictor is fixed. In practice, however, it may be trained on the labeled sample itself, in which case the same labeled observations are used both to fit the predictor and to calibrate its scores. A cross-fitted version avoids this reuse by replacing the single prediction function with out-of-fold predictions and then fitting the calibrator to the resulting labeled out-of-fold scores.

1. Partition the labeled sample indices into folds $I_1, \ldots, I_K$.
2. For each fold k, train the black-box regression $m^{(-k)}$ on the labeled observations outside I_k, compute out-of-fold predictions $m^{(-k)}(X_i)$ for $i \in I_k$, and compute predictions $m^{(-k)}(\widetilde{X}_j)$ on the unlabeled sample.
3. Pool the labeled out-of-fold predictions $\{m^{(-k(i))}(X_i)\}_{i=1}^{n}$ and fit a calibrator f_n to the labeled pairs $(m^{(-k(i))}(X_i), Y_i)$.
4. Form calibrated predictions
$$m^{\star}_n(X_i) = f_n(m^{(-k(i))}(X_i))$$
and
$$m^{\star}_n(\widetilde{X}_j) = \frac{1}{K}\sum_{k=1}^{K} f_n(m^{(-k)}(\widetilde{X}_j)),$$
and then compute the plug-in estimator in equation (3).

This cross-fitted modification is the natural DML-style implementation when one wishes to reduce overfitting from using the same labeled data both to estimate the initial regression and to calibrate it for inference (Zheng and Van Der Laan, 2010; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). Related procedures have been used for debiased inference in van der Laan et al. (2024b,c); Rabenseifner et al. (2025), and a related cross-calibration variant appears in van der Laan et al. (2023).

4 Theory for isotonic-calibrated PPI

We study the theory of isotonic-calibrated PPI through the lens of *calibeating*. Our aim is to understand when isotonic post-processing of an available score can improve on the original score and on simpler calibration rules. We analyze the isotonic-calibrated PPI estimator $\widehat{\psi}_{\text{iso}}$, establishing asymptotic linearity, asymptotic normality, and the validity of Wald inference. We then characterize its efficiency, showing when it is asymptotically at least as efficient as AIPW, PPI, PPI++, and linear calibration, and when it attains the minimum asymptotic variance among regular estimators based on $m(X)$ or on the full covariate vector X (Bickel et al., 1993; Van der Vaart, 2000). Finally, we study the calibration side of calibeating. We show that isotonic calibration is

first-order optimal in a natural sense: once a score has been isotonically calibrated, no further post-processing based only on the fitted score can improve either its mean-squared prediction error or the leading asymptotic behavior of the resulting plug-in estimator. To keep the exposition light, we focus on the non-cross-fitted estimator and treat the initial black-box score m as fixed.

4.1 Asymptotic normality, inference, and efficiency

We let f_0 denote a population isotonic calibration map for the fixed score m, defined by

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \mathbb{E}_{P_0}\big[\{Y - f(m(X))\}^2\big], \qquad m_0 = f_0 \circ m. \tag{15}$$

Thus, m_0 is the population L^2-optimal monotone transformation of the initial score $m(X)$. Because the identity map belongs to $\mathcal{F}_{\text{iso}}$, m_0 has squared-error risk no larger than that of the original score m.

Assumption 2 (Regularity conditions for isotonic calibration)**.** Assume:

(i) **Boundedness:** There exists $C_0 < \infty$ such that $|\mathbb{E}_{P_0}[Y \mid m(X)]| \le C_0$ almost surely and $\sup_x |m^\star_{n,\text{iso}}(x)| = O_p(1)$.

(ii) **Controlled tails:** The conditional error $Y - \mathbb{E}_{P_0}[Y \mid m(X)]$ is sub-Gaussian or subexponential under P_0.

Condition (i) requires only that the regression target be bounded and that the isotonic estimator remain bounded in probability. Condition (ii) allows Y to be unbounded, covering both Gaussian-like tails and somewhat heavier exponential-type tails. In particular, it holds for binary, Gaussian, Poisson, and bounded outcomes. Existing results on isotonic regression under heavy-tailed errors also suggest some robustness to violations of these conditions (Han and Wellner, 2018, 2019).

The target of the asymptotic analysis is the population isotonic limit m_0, which need not equal the true regression μ_0. Under Assumption 2, the isotonic estimator attains the classical rate of $n^{-2/3}$; see, for example, Chatterjee et al. (2015); Yang and Barber (2018).

Theorem 2 (Asymptotic linearity under the two-sample model)**.** *Suppose Assumptions 1 and 2 hold. Then*

$$\|m^\star_{n,\text{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3});$$

Furthermore,

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_0 - \psi_0\} + O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big).\end{aligned} \tag{16}$$

Thus, under the $\sqrt{M}$ scaling, $\widehat{\psi}_{\text{iso}}$ is asymptotically linear with influence pair $(D^L_{m_0}, D^U_{m_0})$ given by

$$D^L_{m_0}(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D^U_{m_0}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0.$$

The next corollary records the resulting asymptotic normality and the asymptotic validity of standard Wald confidence intervals. The limiting variance of $\widehat{\psi}_{\text{iso}}$ is

$$\sigma_0^2 := \rho_0 \operatorname{Var}\{D^L_{m_0}(X,Y)\} + (1-\rho_0)\operatorname{Var}\{D^U_{m_0}(\widetilde{X})\}.$$

Corollary 1 (Asymptotic normality and Wald inference)**.** *Under the conditions of theorem 2, assume that $\sigma_0^2 > 0$. Define the empirical variance*

$$\widehat{\sigma}^2 := \rho_n \mathbb{P}_n^L[(\widehat{D}^L)^2] + (1-\rho_n)\mathbb{P}_N^U[(\widehat{D}^U)^2],$$

where

$$\widehat{D}_i^L := m^\star_{n,\mathrm{iso}}(X_i) - \widehat{\psi}_{\mathrm{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{iso}}(X_i)\}, \qquad \widehat{D}_j^U := m^\star_{n,\mathrm{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{iso}}.$$

Then

$$\frac{\sqrt{M}\,(\widehat{\psi}_{\mathrm{iso}} - \psi_0)}{\widehat{\sigma}} \xrightarrow{d} N(0,1).$$

The next corollary describes two important efficiency regimes: one in which isotonic calibration attains full semiparametric efficiency, and one in which it is optimal among estimators that use the covariates only through the one-dimensional score $m(X)$.

Corollary 2 (Efficiency regimes)**.** *Under the conditions of theorem 2, the following hold.*

(i) If $m_0 = \mu_0$ almost surely, then $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

(ii) If $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone increasing in $m(X)$, then $m_0(X) = \mathbb{E}_{P_0}[Y \mid m(X)]$ almost surely. Hence $\widehat{\psi}_{\mathrm{iso}}$ is efficient for ψ_0 in the reduced two-sample experiment obtained by replacing X with the scalar score $m(X)$.

Remark 5 (Practical takeaway)**.** Isotonic calibration can improve efficiency when the original score is informative but miscalibrated. The resulting estimator behaves as if $m(X)$ were replaced by its best monotone calibration $m_0(X)$. Because the isotonic class contains the identity map and all affine maps with positive slope, the isotonic-calibrated estimator has limiting variance no larger than that of any estimator that is first-order equivalent to AIPW with score $\theta\{m(X)\}$ for such a transformation θ. This includes AIPW and PPI based on $m(X)$, as well as PPI++, and linear calibration whenever the corresponding scaling or slope coefficient is nonnegative. Thus isotonic calibration is asymptotically at least as efficient, at first order, as these methods, and can be strictly more efficient when the best monotone calibration is genuinely nonlinear. More generally, when $\mathbb{E}_{P_0}[Y \mid m(X)]$ is monotone, isotonic calibration already attains the best first-order performance available from monotone post-processing of $m(X)$.

Remark 6 (When full efficiency is attained)**.** Full efficiency requires that the calibrated target recover all regression information relevant to ψ_0. This holds if $m_0 = \mu_0$ almost surely. More generally, full efficiency fails when the one-dimensional score $m(X)$ omits predictive information in X that cannot be recovered through monotone calibration.

4.2 Calibeating and no further first-order gain from post-processing

We now ask whether the isotonic-calibrated score can be improved by an additional post-processing step. Since $\widehat{\psi}_{\mathrm{iso}}$ depends on the covariates only through the one-dimensional fitted score $m^\star_{n,\mathrm{iso}}(X)$, it is natural to study the reduced two-sample experiment

$$\mathcal{E}_{m^\star_{n,\mathrm{iso}}} := \Big(\{(m^\star_{n,\mathrm{iso}}(X_i), Y_i)\}_{i=1}^n,\ \{m^\star_{n,\mathrm{iso}}(\widetilde{X}_j)\}_{j=1}^N\Big),$$

in which the full covariate X is replaced by the scalar score $m^\star_{n,\mathrm{iso}}(X)$.

Within this reduced experiment, a natural benchmark is the oracle regression function $\mathbb{E}_{P_0}[Y \mid m^\star_{n,\mathrm{iso}}(X)]$. The result below compares $\widehat{\psi}_{\mathrm{iso}}$ to this benchmark and shows that additional post-processing of the fitted score yields no first-order improvement.

Assumption 3 (Bounded-variation score regression)**.** The map $t \mapsto \mathbb{E}_{P_0}[Y \mid m(X) = t]$ has bounded variation on the support of $m(X)$.

Assumption 3 is mild. It requires only that the population regression of Y on the initial score $m(X)$, viewed as a one-dimensional function of the score, have bounded total variation. In particular, it allows jumps and is weaker than differentiability or Lipschitz conditions.

Theorem 3 (No further first-order improvement from post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{\psi}_{\text{eff}} := \widehat{\psi}(\widehat{m}_0), \qquad \widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m^\star_{n,\text{iso}}(X)]$$

denote the benchmark estimator in the reduced experiment based on the isotonic score $m^\star_{n,\text{iso}}(X)$. Then

$$\|\widehat{m}_0 - m^\star_{n,\text{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3}),$$

and

$$\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = o_p(M^{-1/2}).$$

In particular, $\widehat{\psi}_{\text{iso}}$ and $\widehat{\psi}_{\text{eff}}$ have the same asymptotic variance.

Theorem 3 implies that $m^\star_{n,\text{iso}}$ is calibrated in the sense that $\|\widehat{m}_0 - m^\star_{n,\text{iso}}\|^2_{2,P_{0,X}} = O_p(n^{-2/3})$. Theorem 2 further shows that $\|m^\star_{n,\text{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3})$, so this calibration does not asymptotically degrade predictive performance. We therefore say that $m^\star_{n,\text{iso}}$ *prediction calibeats* the original predictor m.

Theorem 3 also shows that the asymptotic variance of $\widehat{\psi}_{\text{iso}}$ matches that of the oracle estimator $\widehat{\psi}_{\text{eff}}$, which uses the oracle score $\widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m^\star_{n,\text{iso}}(X)]$. Equivalently, within the reduced experiment based on $m^\star_{n,\text{iso}}(X)$, $\widehat{m}_0(X)$ is the oracle post-processing that attains the efficiency bound. Moreover, Theorem 2 shows that $\widehat{\psi}_{\text{iso}}$ achieves the optimal limiting variance within the class of monotone post-processings, and is therefore at least as efficient as AIPW based on the original score $m(X)$. Hence, $\widehat{\psi}_{\text{iso}}$ *functional-estimation calibeats* $\widehat{\psi}_{\text{AIPW}}$ and $\widehat{\psi}_{\text{PPI}}$. In practical terms, once the score has been isotonic-calibrated, there is no first-order benefit to further post-processing: it does not improve prediction asymptotically, and it does not improve the efficiency of the resulting estimator.

5 Experiments

Across all experiments we use the same comparison hierarchy. The labeled-only estimator (reported as Naive in the simulation) is the safety baseline. AIPW is the main raw-score baseline, with PPI as the weaker comparator. PPI++ and AIPW-EM are the efficiency-rescaling competitors, and `LinearCal`, `AutoCal`, and related calibration rules are the proposed fix when the score is miscalibrated. The three subsections below therefore move from a controlled mechanism check, to reproduced benchmark evidence, to a practical LLM-evaluation setting.

5.1 Simulation study

We begin with a controlled binary-outcome simulation designed to isolate score miscalibration. Let $S \sim N(0,1)$, define

$$\mu_0(S) = \{1 + \exp(-5S)\}^{-1},$$

and draw $Y \mid S \sim \text{Bernoulli}\{\mu_0(S)\}$. To induce substantial miscalibration, we apply an affine-biased monotone distortion on the logit scale,

$$m(X) = \text{clip}\left[-0.15 + 0.75\,\sigma\big\{0.8\,z(S) + 0.1\,z(S)^3 - 1\big\},\, 0.01,\, 0.99\right],$$

where $z(S) := \text{logit}\{\mu_0(S)\}$. This transformation preserves ranking information while introducing nonlinear distortion on the probability scale. We compare the labeled-only mean (Naive), PPI, AIPW, PPI++, an unclipped efficiency-maximized AIPW benchmark (AIPW-EM), linear calibration (LinearCal), monotone spline calibration (MonoSpline) (Ramsay, 1988; Jiang et al., 2011), isotonic calibration (IsoCal), and an adaptive selector (AutoCal) that chooses among AIPW and the main calibration rules by cross-validated empirical efficiency. Here PPI++ is the official `ppi_py` implementation, while AIPW-EM uses the same one-dimensional empirical efficiency-maximization idea without clipping the selected scaling coefficient. The

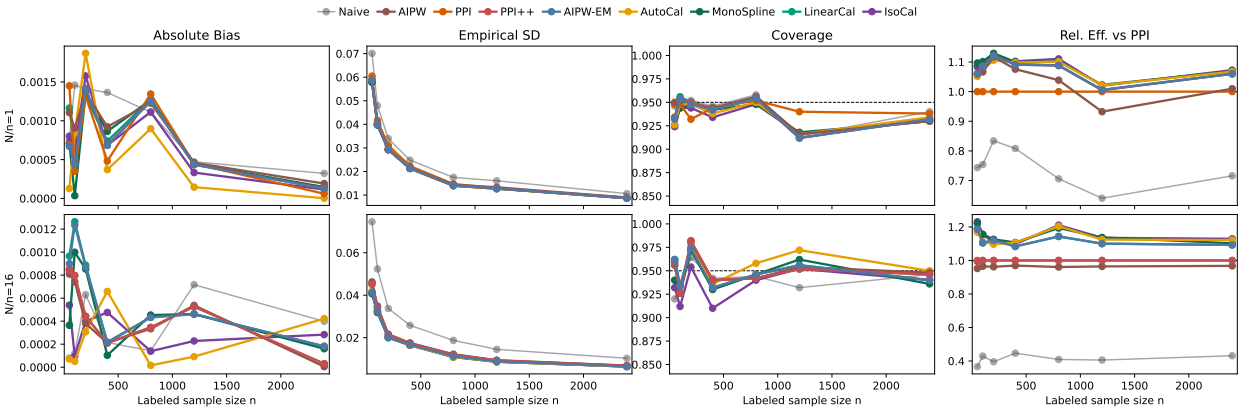

Figure 2: Synthetic simulation study with a poorly calibrated score. The top row shows the balanced case $N/n = 1$, and the bottom row shows the large-unlabeled case $N/n = 16$. The four columns report absolute bias, empirical Monte Carlo standard deviation, coverage, and relative efficiency versus PPI.

labeled sample size ranges over $n \in \{50, 100, 200, 400, 800, 1200, 2400\}$, with $N/n = 1$ and $N/n = 16$. All figures and summaries are based on 500 Monte Carlo repetitions.

figure 2 isolates the mechanism. All score-based AIPW-type estimators are essentially unbiased, so the main comparison is efficiency: AIPW improves on PPI, and calibration helps when the score is genuinely miscalibrated. At $n = 400$ and $N/n = 16$, the calibrated methods reduce RMSE from about 0.0172 for PPI to about 0.0164; by $n = 1200$ in the same regime, RMSE is about 0.0087 for AutoCal, MonoSpline, and IsoCal, compared with 0.0092 for PPI. The PPI++ comparison is most revealing when the unlabeled sample is large: AIPW-EM stays essentially on top of `LinearCal` while clipped PPI++ falls back toward plain PPI. Averaging over the poorly calibrated $N/n = 16$ designs, RMSE is about 0.0196 for AIPW-EM and `LinearCal`, versus 0.0208 for PPI++, reflecting the fact that the official `ppi_py` implementation clips the optimized scaling coefficient to $[0, 1]$ while AIPW-EM does not. AutoCal stays close to the best fixed score-based option, and `MonoSpline` offers a smooth monotone alternative that is usually close to isotonic calibration.

5.2 Empirical Illustration

We evaluate the proposed calibration-based estimators on the mean-estimation benchmarks used in the original experiments of Angelopoulos et al. (2023a,b), as provided in the `ppi_py` package. We use the same `ppi_py` datasets and labeled-sample-size grids, repeatedly split each fully labeled benchmark into labeled and unlabeled subsets, evaluate all estimators on the same split, and treat the full-sample mean as ground truth.[5] The current draft summaries average over 500 random splits at each n. We report the official `forest`, `galaxies`, and `census_income` benchmarks and focus the main text on the primary comparator set already shown in the figure: PPI, AIPW, PPI++, AIPW-EM, `LinearCal`, and `AutoCal`. Appendix L gives the fuller method inventory, calibration-specific comparisons, and dataset-by-regime summary.

Figure 3 shows the same pattern on the reproduced benchmarks. AIPW is the stronger raw-score baseline, with PPI competitive only when the unlabeled sample is extremely large. The clearest calibration win appears in `census_income`, where `LinearCal` performs best on average and the appendix comparison in figure 5 shows that `IsoCal` and `MonoSpline` remain broadly competitive without changing the main ranking. The added efficiency-maximization baseline sharpens the comparison with PPI++: across the 30 dataset-size cells in figure 3, AIPW-EM has lower MSE than clipped PPI++ in 29 and is nearly tied with `LinearCal`;

[5] Because repeated sample splits of a fixed finite benchmark dataset are not i.i.d. draws from a superpopulation, asymptotic Wald confidence intervals can exhibit either undercoverage or overcoverage. This is not specific to our calibration extensions, but is a feature of evaluating Wald-type intervals under finite-population resampling.

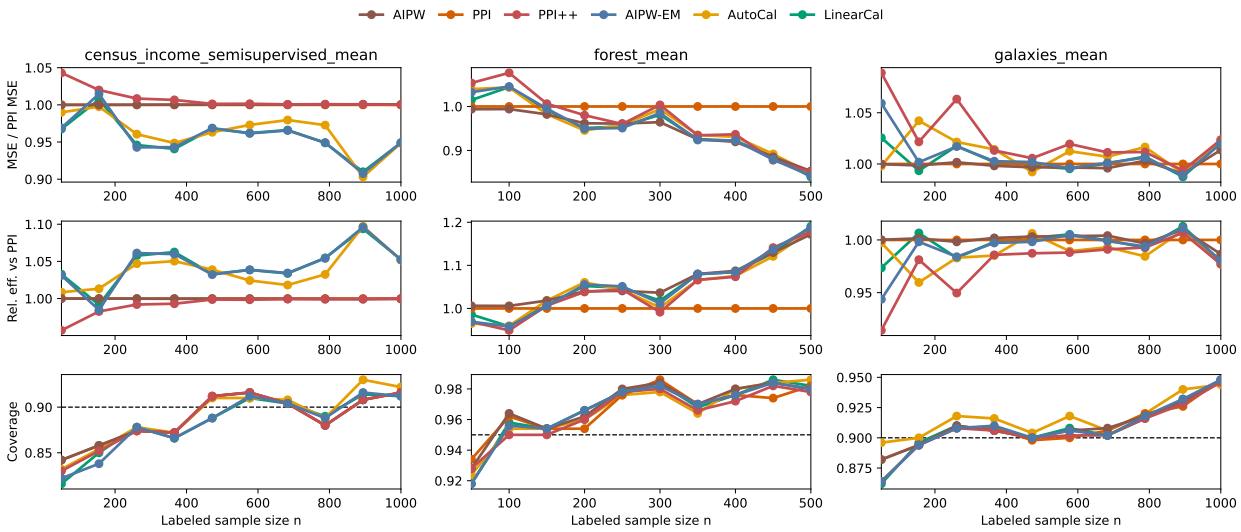

Figure 3: Main-text benchmark summary for the reproduced `ppi_py` datasets. Panels report normalized MSE relative to PPI, relative efficiency versus PPI, and coverage for the main comparator set.

averaged across all cells, mean MSE is about 4.75×10^6 for AIPW-EM, 4.75×10^6 for `LinearCal`, and 5.03×10^6 for PPI++, with essentially identical coverage near 0.919. This mirrors the simulations: once the empirical efficiency-maximization step is left unclipped, its finite-sample behavior moves much closer to `LinearCal`. `AutoCal` acts as a practical hedge that tends to stay close to the best fixed score-based method. Appendix L gives the fuller dataset-by-regime summary.

5.3 LLM evaluation benchmark

We now turn to a practical LLM-evaluation setting where the same comparison hierarchy is applied to real public model outputs, public evaluator scores, and public labels. The first track uses PPE Human Preference V1, where the estimand for each target model is its human win share across prompt–response comparisons, with ties counted as half-wins. The second uses a PPE correctness suite over MMLU-Pro, MATH, GPQA, IFEval, and MBPP+, where each unit compares a sampled conflict pair from the same generator model and the estimand is the correctness rate of the first answer in the pair. In both tracks the prompts, outputs, judgments, and evaluator scores are real; only the labeled/unlabeled split is rerandomized. Across 200 repeated splits we compare labeled-only estimation against raw PPI/AIPW, the efficiency-maximized variants, and the calibration-based estimators from our main experiments. The main-text summary in Figure 4 and table 1 focuses on these two PPE tracks.

Two messages stand out. First, raw proxy use can fail badly. Raw AIPW is again the stronger raw-score baseline, but even AIPW can be dramatically worse than labeled-only when the reward-model margin is badly scaled. On PPE Human at $n = 100$, raw PPI and raw AIPW have MSE 0.495 and 0.440, compared with 0.001425 for labeled-only and 0.001391 for `LinearCal`; on PPE Correctness at $n = 100$, the corresponding MSEs are 0.534, 0.353, 0.002010, and 0.001813. These estimators remain valid first-order corrections, but the extra proxy term can inflate variance enough that labeled-only is far safer.

Second, calibration makes the same evaluator scores practically useful. `LinearCal` is the strongest and most stable method overall, with `AutoCal` close behind. On PPE Human, `LinearCal` is best from $n = 50$ onward, improving on labeled-only from 0.001425 to 0.001391 at $n = 100$ and from 0.000286 to 0.000278 at $n = 400$, corresponding to modest 2.5% to 3.1% label savings. On PPE Correctness, the gains are clearer: `LinearCal` is best at every budget, improving from 0.002010 to 0.001813 at $n = 100$ and from 0.000248 to 0.000222 at $n = 400$, with about 7.5% to 10.5% label savings. The more flexible calibration rules are also competitive once the labeled sample is moderately large: on PPE Human, `AutoCal` and `MonoSpline` remain in the 2.5%

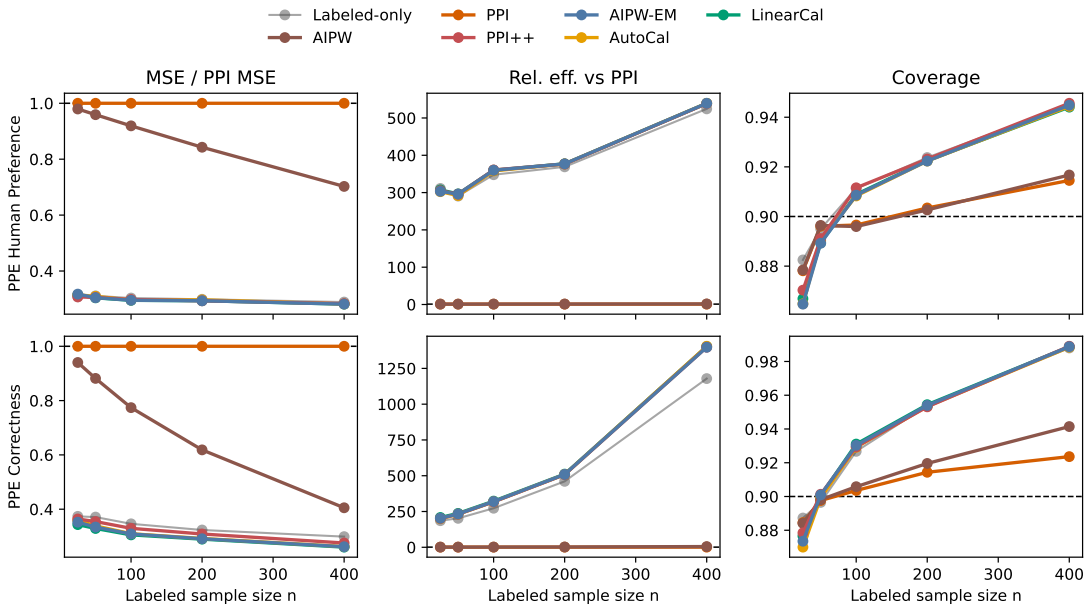

Figure 4: PPE-centered LLM-evaluation benchmark. Rows correspond to PPE Human Preference and PPE Correctness, while columns report normalized MSE relative to PPI, relative efficiency versus labeled-only, and coverage after macro-averaging across the public PPE evaluator models.

to 2.8% label-savings range from $n = 100$ to $n = 400$; on PPE Correctness, they remain in roughly the 9% to 10% range over the same budgets. `MonoSpline` therefore substantially improves over raw PPI/AIPW and is usually competitive with PPI++ and AIPW-EM, although it remains slightly less stable than `LinearCal`.

Third, the evaluator-specific appendix breakdown (Figure 7) shows that the most extreme raw-proxy failures are driven by evaluators such as Skywork and Athene, whereas ArmoRM is much more benign. Finally, calibration also helps on the downstream ranking task induced by PPE Human. Raw PPI/AIPW recover the model ordering poorly, whereas calibrated and efficiency-maximized methods recover it well. At $n = 200$, `AutoCal` attains the highest average Spearman correlation, about 0.905; at $n = 400$, `LinearCal` gives the best overall ranking quality, with Spearman about 0.941 and top-1 identification about 0.646. We therefore view calibration not as a marginal efficiency tweak, but as the step that makes cheap public LLM evaluators safe enough to use in practice; see Appendix L.2.

6 Conclusion and closing remarks

We introduced *Calibrated Prediction-Powered Inference*, a simple framework for semisupervised mean estimation based on three steps: fit a prediction score, calibrate it on the labeled sample, and average the calibrated predictions over the pooled covariate sample. This simple plug-in procedure fits naturally within the broader literature on semiparametric debiasing and can be viewed as a semisupervised specialization of calibrated debiased machine learning (van der Laan et al., 2024c).

Our results highlight a simple message. Calibration can improve semisupervised efficiency without retraining the original model, and isotonic calibration can *calibeat* the original score and simpler post-processing rules at first order. At the same time, once the score has been isotonicly calibrated, further post-processing of that fitted score brings no additional first-order gain. Empirically, this suggests a transparent and practical workflow: when the original score is already well calibrated, calibration changes little, whereas under meaningful miscalibration, even simple linear or monotone calibration can improve efficiency while preserving a transparent plug-in form.

n	Estimator	Human MSE	Human label savings	Correctness MSE	Correctness label savings	Avg. cov.
100	Labeled-only	0.001425	0.0%	0.002010	0.0%	0.919
100	PPI	0.495213	0.4%	0.533854	1.0%	0.900
100	AIPW	0.440144	0.4%	0.352885	0.9%	0.901
100	PPI++	0.001395	2.7%	0.001875	7.0%	0.921
100	AIPW-EM	0.001392	3.1%	0.001826	9.5%	0.919
100	LinearCal	0.001391	3.1%	0.001813	10.1%	0.920
100	AutoCal	0.001405	2.8%	0.001835	9.4%	0.919
100	MonoSpline	0.001412	2.5%	0.001839	9.2%	0.915
400	Labeled-only	0.000286	0.0%	0.000248	0.0%	0.966
400	PPI	0.147604	0.5%	0.272679	2.0%	0.919
400	AIPW	0.087701	0.4%	0.042829	0.9%	0.929
400	PPI++	0.000279	2.5%	0.000229	8.3%	0.967
400	AIPW-EM	0.000278	2.8%	0.000223	10.3%	0.967
400	LinearCal	0.000278	2.8%	0.000222	10.5%	0.966
400	AutoCal	0.000279	2.7%	0.000223	10.2%	0.966
400	MonoSpline	0.000279	2.7%	0.000227	9.4%	0.965

Table 1: PPE-only numerical summary at $n \in \{100, 400\}$, chosen to contrast the moderate-label and larger-label regimes. "Human label savings" and "Correctness label savings" report the estimated fraction of labels saved relative to labeled-only for the same variance; Avg. cov. averages coverage across the two PPE tracks.

Positioning PPI within existing semiparametric methods. More broadly, our results clarify that recent prediction-powered inference methods fit naturally within the standard semiparametric, debiased machine-learning, and flexible covariate-adjustment toolkit for randomized trials and missing-at-random settings. In particular, we show that PPI++ is AIPW with empirical efficiency maximization (Rubin and van der Laan, 2008) and is first-order equivalent to linear calibration and classical prognostic-score regression adjustment (Hansen, 2008; Lin, 2013; Schuler et al., 2022). It is therefore closely related to randomized-trial approaches that learn prognostic scores from larger historical, external-control, or auxiliary data sets and then use them for regression adjustment in smaller trials (Hansen, 2008; Moore and van der Laan, 2009a; Rosenblum and van der Laan, 2009; Benkeser et al., 2021; Højbjerre-Frandsen et al., 2025). This perspective suggests that semisupervised inference may benefit from closer integration with the broader missing-data, causal-inference, and debiased machine-learning literatures. These lines of work address closely related versions of the same underlying problem, so further progress may come from translational work across them, especially in adapting tools developed for randomized trials and missing-at-random settings to semisupervised problems (Petersen and van der Laan, 2014; Dang et al., 2023; Ho et al., 2023; Smith et al., 2023; Ho et al., 2024).

Beyond mean estimation. These connections extend well beyond the semisupervised mean-estimation setting considered here. For example, when missingness affects both outcomes and covariates and may depend on observed covariates and outcomes, as well as in extensions to right-censored time-to-event outcomes and longitudinal settings, existing semiparametric methods can often be applied directly (Robins et al., 1994; van der Laan and Robins, 2003; Bang and Robins, 2005; Moore and van der Laan, 2009b; Rose and van der Laan, 2011; Van der Laan and Rose, 2018). More generally, M-estimation and estimating-equation methods for missing-at-random settings are also well developed (Robins et al., 1994; van der Laan and Robins, 2003; Chernozhukov et al., 2018a). Finally, when outcomes are not missing completely at random, naively pooling labeled and unlabeled data can bias estimation because the missingness mechanism may remain confounded by unobserved outcomes. There is substantial work on combining gold-standard and potentially biased auxiliary data more safely using semiparametric and machine-learning tools (Kallus et al., 2018; Rosenman, 2025; Dang et al., 2025; van der Laan et al., 2026). Similarly, when the unlabeled sample is selected in a data-dependent way, the problem becomes closely related to adaptive and selectively sampled designs, including two-phase and informative sampling (van der Laan, 2008; Chow and Chang, 2008; Rose and van der Laan, 2011; Malenica et al., 2021; Zrnic and Candès, 2024b; Zhang and van der Laan, 2025; Liu et al., 2025). Extending these ideas to semisupervised settings is therefore a natural direction for future work.

Conditional Prediction-Powered Inference There is already a well-developed literature on machine-

learning-assisted estimation of local and conditional regression targets, including regression functions, dose-response curves, and conditional average treatment effects, which can be used to extend inference beyond marginal mean estimation (Rubin and van der Laan, 2006; Díaz and van der Laan, 2013; Kennedy et al., 2017; Bibaut and van der Laan, 2017; van der Laan et al., 2018; Westling et al., 2020; Chernozhukov et al., 2022b; Kennedy, 2023; Foster and Syrgkanis, 2023; Chernozhukov et al., 2023; Luedtke and Chung, 2024; Chernozhukov et al., 2024; Butzin-Dozier et al., 2024; Zhang et al., 2025). In particular, kernel-regression-based (Bibaut and van der Laan, 2017) and isotonic-regression-based (Westling and Carone, 2020) inference can be used for conditional PPI with AIPW-score-based pseudo-outcomes, as in Van Der Laan and Dudoit (2003); Rubin and van der Laan (2006); Kennedy et al. (2017), or more generally through influence functions and Neyman-orthogonal losses (Bibaut and van der Laan, 2017; Chernozhukov et al., 2024). The key idea is that locally smoothed conditional targets are amenable to influence-function-based bias correction, after which inference follows from a triangular-array central limit theorem applied to estimators of a sequence of approximating parameters (Bibaut and van der Laan, 2017; Chernozhukov et al., 2022b; Luedtke and Chung, 2024; Chernozhukov et al., 2024). In the prediction-powered setting, a natural approach is to apply kernel regression to AIPW-style pseudo-outcomes for missing-data problems, following Rubin and van der Laan (2006), Takatsu and Westling (2025), and Bibaut and van der Laan (2017); Kennedy et al. (2017); Chernozhukov et al. (2024).

Calibration for general loss functions. While we focus on calibration of regression functions and inference on means, the ideas here may extend to broader supervised learning problems defined through loss minimization, such as median and quantile regression (Rosenblum and van der Laan, 2009; Noarov and Roth, 2023; Jung et al., 2021; Roth, 2022; Whitehouse et al., 2024). They also extend to continuous linear functionals of a regression function or M-estimand through calibrated DML (van der Laan et al., 2024c, 2025a). More generally, when the target parameter is defined by a moment equation, as in M-estimation, one could debias or calibrate the associated augmented moment equation (van der Laan and Robins, 2003; Chernozhukov et al., 2018a); see Appendix E and Ji et al. (2025). A practical challenge in this setting is that the augmented loss may be nonconvex or may require separate debiasing at each parameter value, which can complicate optimization. In such cases, a more practical alternative may be a multiaccuracy-style adjustment rather than explicit augmentation or calibration, as in the Efficient Plug-in Learning framework of van der Laan et al. (2024a), which addresses this issue for nonconvex Neyman-orthogonal loss functions (Foster and Syrgkanis, 2023).

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Contents

1 Introduction **1**

1.1 Related work 4

2 Two-Sample Semisupervised Setup **6**
2.1 Data structure and notation . . . 6
2.2 Review of AIPW, PPI, and semiparametric efficiency . . . 7

3 Calibrated Prediction-Powered Inference **9**
3.1 General estimator and exact AIPW representation . . . 9
3.2 Isotonic-calibrated PPI . . . 11
3.3 Linearly calibrated PPI . . . 12
3.4 Cross-Fitting and Cross-Calibration . . . 14

4 Theory for isotonic-calibrated PPI **14**
4.1 Asymptotic normality, inference, and efficiency . . . 15
4.2 Calibeating and no further first-order gain from post-processing . . . 16

5 Experiments **17**
5.1 Simulation study . . . 17
5.2 Empirical Illustration . . . 18
5.3 LLM evaluation benchmark . . . 19

6 Conclusion and closing remarks **20**

A Additional background on semiparametric statistics **33**

B Constructing confidence intervals via the bootstrap **34**

C Additional remarks **35**

D Prediction-Powered Causal Inference via Calibration (i.e., Calibrated DML) **36**

E General Moment Equations Under Missing at Random **38**

F Proof of the representation theorem **41**

G Proofs of first-order equivalence between PPI++ and linear calibration **42**

H Proof of Proposition 1 **44**

I Proof of asymptotic linearity **53**

J Proof of post-processing equivalence and reduced-model efficiency **54**

K Pooled i.i.d. Missing-Data Formulation **55**

L Additional empirical details **55**

L.1 LLM evaluator breakdown 57

L.2 LLM ranking supplement 57

M Python Code **57**

M.1 Isotonic regression via XGBoost 57

M.2 Isotonic-calibrated plug-in 59

M.3 Linear calibration 62

A Additional background on semiparametric statistics

The next result shows that the efficient choice in Proposition 1 is not merely optimal within the AIPW class. In fact, every regular estimator[6] in the unrestricted two-sample model is asymptotically equivalent to $\widehat{\psi}(f)$ for some score function f. Thus, the AIPW family characterizes the full class of regular estimators at first order in this model.

Proposition 5 (Characterization of regular estimators)**.** *Under Assumption 1, a mean-zero pair (D^L, D^U) is the influence pair of a regular estimator of ψ_0 in the unrestricted two-sample model if and only if there exists a function $f_0 : \mathcal{X} \to \mathbb{R}$ with finite second moment, potentially depending on P_0, such that*

$$(D^L, D^U) = (D^L_{f_0}, D^U_{f_0}),$$

where D^L_f and D^U_f are given in Proposition 1. Equivalently, for every regular estimator $\widehat{\psi}^\dagger$, there exists such an f_0 for which

$$\widehat{\psi}^\dagger - \widehat{\psi}(f_0) = o_p(M^{-1/2}).$$

This characterization shows that the AIPW class is the relevant first-order benchmark in this model. The next remark illustrates that several estimators not always written in AIPW form—including regression adjustment, balancing-weight adjustment, and TMLE—can nonetheless be viewed as finite-sample or asymptotic equivalents of AIPW estimators (Rotnitzky et al., 2025; Van Der Laan and Rubin, 2006), while sometimes offering better finite-sample performance (Porter et al., 2011; Jin, 2026; van der Laan et al., 2025a).

Remark 7 (Plug-in approaches to bias correction and TMLE)**. Regression plug-in estimators.** A particularly simple member of this first-order class is obtained through plug-in regression adjustment (Scharfstein et al., 1999; Van Der Laan and Rubin, 2006; Rosenblum and van der Laan, 2009; Rotnitzky et al., 2025). Let

$$m_n(X) = \hat{a}_{\text{lin}}\, m(X) + \hat{b}_{\text{lin}}, \qquad (\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) = \arg\min_{a,b} \sum_{i=1}^{n} \{Y_i - a\, m(X_i) - b\}^2.$$

The corresponding plug-in estimator is

$$\rho_n \mathbb{P}^L_n\{m_n(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m_n(\widetilde{X})\}.$$

For any $\lambda \in \mathbb{R}$, this estimator also admits the AIPW representation

$$\rho_n \mathbb{P}^L_n\{m_n(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m_n(\widetilde{X})\} + \lambda \mathbb{P}^L_n\big[Y - m_n(X)\big],$$

because the augmentation term $\mathbb{P}^L_n[Y - m_n(X)]$ is zero by the least-squares normal equations. Thus, when m_n converges to the best linear predictor of Y given $m(X)$, this plug-in estimator is typically at least as efficient as $\widehat{\psi}_{\text{AIPW}}$ and $\widehat{\psi}_{\text{PPI}}$, since it corresponds to an influence function with smaller or equal variance.

[6] Informally, a regular estimator is one whose large-sample behavior remains stable under small changes in the data-generating distribution (Van der Vaart, 2000).

Balancing-weight estimators. The same estimator can also be written in terms of balancing weights. Let $w_n(X) = \hat{a}_{\text{bal}} + \hat{b}_{\text{bal}}\, m(X)$ be affine weights that balance $(1, m(X))$, in the sense that

$$\mathbb{P}_n^L\{w_n(X)\} = 1 \quad \text{and} \quad \mathbb{P}_n^L\big[m(X)w_n(X)\big] = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\}.$$

Then the corresponding balancing-weight estimator satisfies

$$\mathbb{P}_n^L\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\big[w_n(X)\{Y - m(X)\}\big].$$

In fact, the plug-in and balancing-weight estimators are algebraically equivalent (Chattopadhyay and Zubizarreta, 2023; Bruns-Smith et al., 2025):

$$\mathbb{P}_n^L\big\{w_n(X)Y\big\} = \rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\}.$$

TMLE. More broadly, such debiased plug-in estimators can be viewed as instances of TMLE (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011; Højbjerre-Frandsen and Schuler, 2026), a general framework for constructing plug-in estimators that are debiased and, in many cases, efficient by updating nuisance components to remove first-order bias. In the present setting, either the weights or the regression model are updated so that the corresponding augmentation term vanishes. In weighting-based TMLE, the update balances on the prediction model $m(X)$ (e.g., Hejazi and van der Laan 2023); one may either include the known weight $1/\rho_n$ as an offset or balance on the intercept as well. In regression-based TMLE, the prediction model $m(X)$ may be included either as a covariate or as an offset (e.g., Lendle et al. 2015), and the intercept adjustment accounts for the known inverse missingness-probability weight $1/\rho_n$. The specific targeting step is ultimately a practical choice, provided it solves the relevant estimating equation and removes the augmentation term; different choices may also yield additional benefits, such as further bias reduction or doubly robust inference (Van der Laan, 2014; Carone et al., 2014; Díaz and van der Laan, 2017; Benkeser et al., 2017; van der Laan et al., 2021, 2024c).

B Constructing confidence intervals via the bootstrap

Confidence intervals may be constructed using the standard Wald approximation from Corollary 1, namely

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\, \frac{\hat{\sigma}}{\sqrt{n+N}}.$$

As an alternative, one may use the nonparametric bootstrap and refit the isotonic calibration step within each bootstrap replicate, as proposed and theoretically studied by van der Laan et al. (2024c); see also Tang and Westling (2024). This avoids direct estimation of the influence-function variance and automatically propagates uncertainty from estimation of the calibration map.

For each bootstrap replicate $b = 1, \ldots, B$, we sample with replacement from the labeled observations $\{(X_i, Y_i)\}_{i=1}^n$ and, independently, from the unlabeled observations $\{\widetilde{X}_j\}_{j=1}^N$, obtaining bootstrap samples

$$\{(X_i^{*(b)}, Y_i^{*(b)})\}_{i=1}^n, \qquad \{\widetilde{X}_j^{*(b)}\}_{j=1}^N.$$

We then refit the isotonic calibrator on the bootstrap labeled sample:

$$\hat{f}^{*(b)} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i^{*(b)} - f(m(X_i^{*(b)}))\big\}^2,$$

and define

$$m_{\text{iso}}^{*(b)}(X) := \hat{f}^{*(b)}\{m(X)\}.$$

The corresponding bootstrap replicate of the estimator is

$$\widehat{\psi}_{\text{iso}}^{*(b)} := \frac{1}{n+N}\left\{\sum_{i=1}^n m_{\text{iso}}^{*(b)}(X_i^{*(b)}) + \sum_{j=1}^N m_{\text{iso}}^{*(b)}(\widetilde{X}_j^{*(b)})\right\}.$$

After computing B bootstrap replicates, one may form a percentile interval using the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of

$$\{\widehat{\psi}_{\text{iso}}^{*(b)}\}_{b=1}^{B}.$$

Alternatively, one may use the bootstrap standard error

$$\widehat{\sigma}_{\text{boot}}^2 := \frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\psi}_{\text{iso}}^{*(b)} - \bar{\psi}_{\text{iso}}^{\star}\right)^2, \qquad \bar{\psi}_{\text{iso}}^{\star} := \frac{1}{B}\sum_{b=1}^{B}\widehat{\psi}_{\text{iso}}^{*(b)},$$

and report the normal-approximation interval

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\,\widehat{\sigma}_{\text{boot}}.$$

The key implementation detail is that the isotonic regression step must be refit within each bootstrap replicate rather than held fixed. Otherwise, the resulting interval does not reflect uncertainty from estimation of the calibration map.

C Additional remarks

Remark 8 (Empirical efficiency maximization as a learning objective)**.** We review Rubin and van der Laan (2008). The empirical variance criterion from Section 2.2,

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L\left[\left\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\right\}^2\right] + (1-\rho_n)\mathbb{P}_N^U\left[\left\{f(\widetilde{X}) - \widehat{\psi}(f)\right\}^2\right],$$

is not tied to the one-dimensional scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$. For any candidate class $\mathcal{G} \subset L^2(P_{0,X})$, one may instead define

$$\hat{f} \in \arg\min_{f\in\mathcal{G}} \sigma_n^2(f), \qquad \widehat{\psi}_{\text{eem}} := \widehat{\psi}(\hat{f}),$$

so empirical efficiency maximization learns the augmentation $f(X)$ directly.

At the population level, let

$$V(f) := \rho_0 \mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\}$$

denote the asymptotic variance of $\widehat{\psi}(f)$. Using Proposition 1,

$$V(f) = \mathrm{Var}(Y) + \frac{1-\rho_0}{\rho_0}\mathrm{Var}\{Y - f(X)\}.$$

Equivalently, with $\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0$,

$$V(f) = \mathrm{Var}(Y) + \frac{1-\rho_0}{\rho_0}\mathbb{E}\big[(Y - \tilde{f}(X))^2\big].$$

Thus the efficiency-optimal choice is the $L^2(P_{0,X})$-projection of μ_0 onto the normalized class $\{\tilde{f} : f \in \mathcal{G}\}$. If $\mathcal{G}$ is closed under additive constants, then minimizing $V(f)$ is equivalent to ordinary least-squares regression of Y on X over $\mathcal{G}$. In particular, rather than optimizing the empirical efficiency criterion $\sigma_n^2(f)$ directly, one can equivalently optimize the usual empirical mean-squared-error objective on the labeled sample.

This makes richer efficiency-maximizing classes immediate. For example, if $\mathcal{G} = \{b + h : h \in \mathcal{H},\ b \in \mathbb{R}\}$ for an RKHS $\mathcal{H}$ with kernel K, then a penalized version of empirical efficiency maximization is

$$(\hat{b}_\lambda, \hat{h}_\lambda) \in \arg\min_{b\in\mathbb{R},\ h\in\mathcal{H}} \left\{\mathbb{P}_n^L\big[(Y - b - h(X))^2\big] + \lambda\|h\|_{\mathcal{H}}^2\right\},$$

which is just kernel ridge regression on the labeled sample, up to the irrelevant additive constant in the variance criterion. By the representer theorem,

$$\hat{h}_\lambda(\cdot) = \sum_{i=1}^{n} \hat{\alpha}_i K(X_i, \cdot),$$

and, after centering the kernel or including the intercept explicitly,

$$\hat{\alpha} = (K_n + n\lambda I_n)^{-1}(Y - \hat{b}_\lambda \mathbf{1}_n).$$

Plugging $\hat{f}_\lambda = \hat{b}_\lambda + \hat{h}_\lambda$ into $\widehat{\psi}(\hat{f}_\lambda)$ yields the corresponding efficiency-maximized AIPW estimator.

For tuning and class selection, it is preferable to evaluate the same criterion out of fold. One may split the labeled sample into folds, fit f^{-k} on the training part of fold k, compute the held-out criterion $\sigma^2_{n,k}(f^{-k})$, and average over folds. The selected class or tuning parameter can then be used in a cross-fitted or cross-averaged AIPW estimator. This gives a direct cross-validated empirical efficiency objective for choosing among classes such as linear, spline, isotonic, or RKHS-based regressors while preserving the usual sample-splitting protection against in-sample overfitting.

Remark 9 (Shrinkage via Venn–Abers)**.** When the labeled sample is small, isotonic calibration may be unstable. A natural alternative is Venn–Abers calibration, originally introduced for binary classification (Vovk and Petej, 2012); see van der Laan and Alaa (2024, 2025) for generalized versions beyond the binary setting. Assume $Y \in [0, 1]$, or rescale otherwise. For each x, augment the labeled calibration sample with the hypothetical point $(m(x), y)$ for $y \in \{0, 1\}$, fit the resulting isotonic calibrators $f_n^{(x,0)}$ and $f_n^{(x,1)}$, and form the interval-valued prediction

$$\big[f_n^{(x,0)}\{m(x)\},\ f_n^{(x,1)}\{m(x)\}\big].$$

Following Vovk and Petej (2012), one may convert this interval to a point prediction by shrinking its midpoint toward the (unadjusted) AIPW estimator:

$$m^\star_{n,\mathrm{VA}}(x) := m^\star_{n,\mathrm{mid}}(x) + \big\{f_n^{(x,1)}\{m(x)\} - f_n^{(x,0)}\{m(x)\}\big\}\big\{\widehat{\psi}_{AIPW} - m^\star_{n,\mathrm{mid}}(x)\big\},$$

where

$$m^\star_{n,\mathrm{mid}}(x) := \frac{f_n^{(x,1)}\{m(x)\} + f_n^{(x,0)}\{m(x)\}}{2}.$$

Since the shrinkage magnitude is proportional to the width of the Venn–Abers interval, this construction can be more stable than plain isotonic calibration in small samples.

Remark 10 (Relation between linear calibration, PPI, and prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through flexible covariate adjustment in small-sample settings (Hansen, 2008; Rosenblum and van der Laan, 2009; Lin, 2013; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Proposition 4 shows that the PPI++ estimator admits the same first-order interpretation. Likewise, the PPI estimator $\widehat{\psi}_{\mathrm{PPI}}$ is algebraically equivalent to the unlabeled-sample plug-in estimator $\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$ based on the intercept-only adjustment

$$\hat{a}_{\mathrm{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^\star_{n,\mathrm{const}}(X) := \hat{a}_{\mathrm{const}} + m(X), \tag{17}$$

which calibrates only the mean. By contrast, the standard AIPW estimator $\widehat{\psi}_{\mathrm{AIPW}}$ is equivalent to the pooled-sample mean $\rho_n \mathbb{P}_n^L\{m^\star_{n,\mathrm{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$.

D Prediction-Powered Causal Inference via Calibration (i.e., Calibrated DML)

This appendix records the direct translation of the two-sample missing-outcome setup to randomized experiments. The central point is that the semisupervised problem studied in the main text and two-arm

causal inference in a randomized trial are the same statistical problem once the unobserved potential outcome in one arm is interpreted as the missing outcome. We then illustrate how our calibrated PPI estimators apply in this setting, where they arise as special cases of calibrated DML (van der Laan et al., 2024c).

We use the standard potential-outcomes notation. Each unit has two potential outcomes, $Y(1)$ under treatment and $Y(0)$ under control, and the observed outcome satisfies

$$Y = AY(1) + (1 - A)Y(0),$$

where $A \in \{0, 1\}$ is the treatment assignment. The mean potential outcomes are

$$\mu_1 := \mathbb{E}\{Y(1)\}, \qquad \mu_0 := \mathbb{E}\{Y(0)\},$$

and the average treatment effect is $\tau := \mu_1 - \mu_0$. In a randomized trial, identification follows from the usual causal conditions: consistency and no interference, exchangeability induced by randomization,

$$A \perp (X, Y(1), Y(0)),$$

and positivity, so that each arm has positive assignment probability.

Now partition the observed data into the two arm-specific samples

$$\mathcal{D}_{n,1} := \{(X_i, Y_i) : A_i = 1\}, \qquad \mathcal{D}_{n,0} := \{(\widetilde{X}_j, \widetilde{Y}_j) : A_j = 0\},$$

with sample sizes $n_1 := |\mathcal{D}_{n,1}|$, $n_0 := |\mathcal{D}_{n,0}|$, and $n = n_1 + n_0$. Let $\mathbb{P}_{n,a}$ denote the empirical mean over arm a, and let $\hat{\pi}_a := n_a/n$. To estimate μ_1, one treats $\mathcal{D}_{n,1}$ as the labeled sample for $Y(1)$ and uses the covariates in $\mathcal{D}_{n,0}$ as the unlabeled sample. To estimate μ_0, one reverses the roles of the two arms. Thus each mean potential outcome is a two-sample missing-outcome estimand of exactly the form analyzed in the main text.

For $a \in \{0, 1\}$, let $m_a(X)$ be an arm-specific prediction score fit using only observations in arm a. The standard arm-specific AIPW estimator is

$$\hat{\mu}_a^{\text{AIPW}} := \mathbb{P}_n\{m_a(X)\} + \mathbb{P}_n\left[\frac{\mathbb{1}(A = a)}{\hat{\pi}_a}\{Y - m_a(X)\}\right].$$

For $a \in \{0, 1\}$, let $m^\star_{n,a}(X)$ denote the calibrated version of the score $m_a(X)$. The calibrated estimator of μ_a is then

$$\hat{\mu}_a^{\text{cal}} := \mathbb{P}_n\{m^\star_{n,a}(X)\} = \hat{\pi}_a\, \mathbb{P}_{n,a}\{m^\star_{n,a}(X)\} + (1 - \hat{\pi}_a)\, \mathbb{P}_{n,1-a}\{m^\star_{n,a}(X)\}. \tag{18}$$

This is exactly the pooled plug-in estimator from the two-sample formulation, with one arm contributing outcomes and the other contributing only covariates. The same estimator also has the exact AIPW form

$$\hat{\mu}_a^{\text{cal}} = \mathbb{P}_n\{m^\star_{n,a}(X)\} + \mathbb{P}_n\left[\frac{\mathbb{1}(A = a)}{\hat{\pi}_a}\{Y - m^\star_{n,a}(X)\}\right]. \tag{19}$$

Because

$$\mathbb{P}_n\left[\frac{\mathbb{1}(A = a)}{\hat{\pi}_a}\{Y - m^\star_{n,a}(X)\}\right] = \mathbb{P}_{n,a}\{Y - m^\star_{n,a}(X)\},$$

the residual correction vanishes exactly whenever the calibrator includes the constant score and therefore enforces mean calibration within arm a. This is the randomized-trial analogue of the exact augmented representation in theorem 1.

The average treatment effect is then estimated by

$$\hat{\tau}^{\text{cal}} := \hat{\mu}_1^{\text{cal}} - \hat{\mu}_0^{\text{cal}}. \tag{20}$$

Equivalently,

$$\hat{\tau}^{\text{cal}} = \mathbb{P}_n\{\tilde{m}_1(X) - \tilde{m}_0(X)\} + \mathbb{P}_n\left[\frac{A}{\hat{\pi}_1}\{Y - \tilde{m}_1(X)\} - \frac{1 - A}{\hat{\pi}_0}\{Y - \tilde{m}_0(X)\}\right].$$

Its influence function is the difference of the two arm-specific AIPW influences,

$$D_\tau(O) = \left\{\tilde{m}_1(X) - \mu_1 + \frac{A}{\pi_1}\big(Y - \tilde{m}_1(X)\big)\right\} - \left\{\tilde{m}_0(X) - \mu_0 + \frac{1-A}{\pi_0}\big(Y - \tilde{m}_0(X)\big)\right\}.$$

Accordingly, Wald inference for the ATE may be based on the empirical variance of the estimated influence values divided by n, or on the nonparametric bootstrap refitting the calibration step within each bootstrap sample.

The practical implementation is therefore immediate. One splits the trial into $\mathcal{D}_{n,1}$ and $\mathcal{D}_{n,0}$, treats one arm as labeled and the other as unlabeled when estimating a given mean potential outcome, fits and calibrates an arm-specific score, averages the calibrated predictions over the pooled covariate sample, and finally subtracts the two arm-specific estimates to obtain the ATE. In this way, the semisupervised mean estimators in the main text become direct estimators of mean potential outcomes and treatment effects in randomized experiments.

E General Moment Equations Under Missing at Random

We briefly review the general AIPW class for moment-equation targets under missing at random, following Robins et al. (1994, 1995); Robins and Rotnitzky (1995); van der Laan and Robins (2003). Let

$$O = (X, R, RY),$$

where $R \in \{0,1\}$ indicates whether Y is observed, and suppose

$$R \perp Y \mid X, \qquad \pi_0(X) := \mathbb{P}(R = 1 \mid X) > 0 \text{ a.s.}$$

Let the target $\theta_0 \in \Theta \subseteq \mathbb{R}^d$ be identified by

$$\mathbb{E}_{P_0}[\varphi(Y, \theta_0)] = 0,$$

where $\varphi : \mathcal{Y} \times \Theta \to \mathbb{R}^d$ is a known identifying function. Define

$$A_0 := \partial_\theta \mathbb{E}_{P_0}[\varphi(Y,\theta)]\big|_{\theta=\theta_0},$$

and assume that A_0 is nonsingular.

Theorem 4 (AIPW class for general moment equations under MAR)**.** *For any square-integrable function $m : \mathcal{X} \times \Theta \to \mathbb{R}^d$, define*

$$\psi(O; \theta, m, \pi_0) := m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}. \tag{21}$$

Then

$$\mathbb{E}_{P_0}[\psi(O; \theta_0, m, \pi_0)] = 0.$$

Hence $\psi(O; \theta, m, \pi_0)$ defines an unbiased AIPW estimating function for θ_0, and $\hat{\theta}$ is obtained by solving

$$\mathbb{P}_n \psi(O; \hat{\theta}, m, \pi_0) = 0.$$

Moreover, let $\hat{\theta}$ be a regular $\sqrt{n}$-consistent solution to this equation, and assume that $\theta \mapsto \mathbb{E}_{P_0}[\psi(O; \theta, m, \pi_0)]$ is differentiable at θ_0 with derivative A_0, and that

$$(\mathbb{P}_n - P_0)\big[\psi(O; \hat{\theta}, m, \pi_0) - \psi(O; \theta_0, m, \pi_0)\big] = o_p\big(n^{-1/2} + \|\hat{\theta} - \theta_0\|\big).$$

Then $\hat{\theta}$ has influence function

$$D_m(O) = -A_0^{-1}\psi(O; \theta_0, m, \pi_0) = -A_0^{-1}\left[m(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m(X, \theta_0)\}\right]. \tag{22}$$

Thus, the class of regular Z-estimators arising from (21) is indexed by the choice of augmentation function m. The efficient choice in the full MAR model is

$$m_0(X, \theta_0) := \mathbb{E}_{P_0}[\varphi(Y, \theta_0) \mid X], \tag{23}$$

which yields the efficient influence function

$$D_{\text{eff}}(O) = -A_0^{-1}\left[m_0(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m_0(X, \theta_0)\}\right]. \tag{24}$$

Proof of theorem 4. For any $\theta \in \Theta$,

$$\begin{aligned}
\mathbb{E}_{P_0}\{\psi(O; \theta, m, \pi_0)\} &= \mathbb{E}_{P_0}\left[m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}_{P_0}\left[m(X, \theta) + \frac{\mathbb{E}_{P_0}(R \mid X, Y)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}_{P_0}\left[m(X, \theta) + \frac{\mathbb{E}_{P_0}(R \mid X)}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}\right] \\
&= \mathbb{E}_{P_0}\big[\varphi(Y, \theta)\big],
\end{aligned}$$

where the third line uses $R \perp Y \mid X$. Evaluating at $\theta = \theta_0$ gives

$$\mathbb{E}_{P_0}\{\psi(O; \theta_0, m, \pi_0)\} = \mathbb{E}_{P_0}\big[\varphi(Y, \theta_0)\big] = 0.$$

Let

$$\Psi_m(\theta) := \mathbb{E}_{P_0}\{\psi(O; \theta, m, \pi_0)\}.$$

The previous calculation shows that $\Psi_m(\theta) = \mathbb{E}_{P_0}[\varphi(Y, \theta)]$ for every θ, so

$$\partial_\theta \Psi_m(\theta)\big|_{\theta=\theta_0} = A_0.$$

Because $\hat{\theta}$ solves the empirical estimating equation,

$$0 = \mathbb{P}_n\psi(O; \hat{\theta}, m, \pi_0) = (\mathbb{P}_n - P_0)\psi(O; \theta_0, m, \pi_0) + \{\Psi_m(\hat{\theta}) - \Psi_m(\theta_0)\} + (\mathbb{P}_n - P_0)\big[\psi(O; \hat{\theta}, m, \pi_0) - \psi(O; \theta_0, m, \pi_0)\big].$$

By differentiability of Ψ_m at θ_0,

$$\Psi_m(\hat{\theta}) - \Psi_m(\theta_0) = A_0(\hat{\theta} - \theta_0) + o(\|\hat{\theta} - \theta_0\|).$$

Since $\hat{\theta}$ is $\sqrt{n}$-consistent, the previous display and the stated remainder condition imply

$$A_0(\hat{\theta} - \theta_0) = -(\mathbb{P}_n - P_0)\psi(O; \theta_0, m, \pi_0) + o_p(n^{-1/2}).$$

Multiplying by $-A_0^{-1}$ gives

$$\hat{\theta} - \theta_0 = (\mathbb{P}_n - P_0)D_m(O) + o_p(n^{-1/2}),$$

with D_m given in equation (22). This proves the influence-function representation.

Now define

$$m_0(X, \theta_0) := \mathbb{E}_{P_0}\{\varphi(Y, \theta_0) \mid X\}, \qquad \delta_m(X) := m(X, \theta_0) - m_0(X, \theta_0).$$

Then

$$D_m(O) = D_{\text{eff}}(O) - A_0^{-1}\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X).$$

Fix any $u \in \mathbb{R}^d$. Because $\delta_m(X)$ is measurable with respect to X,

$$\mathbb{E}_{P_0}\left[\left(1 - \frac{R}{\pi_0(X)}\right)\delta_m(X) \,\middle|\, X\right] = 0.$$

Also, using $R \perp Y \mid X$ and $\mathbb{E}_{P_0}[\varphi(Y,\theta_0) - m_0(X,\theta_0) \mid X] = 0$,

$$\mathbb{E}_{P_0}\left[\frac{R}{\pi_0(X)}\left(1 - \frac{R}{\pi_0(X)}\right)\{\varphi(Y,\theta_0) - m_0(X,\theta_0)\}\middle| X\right] = 0.$$

Therefore, the cross-term vanishes:

$$\mathbb{E}_{P_0}\left[\{u^\top D_{\text{eff}}(O)\}\left(1 - \frac{R}{\pi_0(X)}\right)\{u^\top A_0^{-1}\delta_m(X)\}\right] = 0.$$

Moreover,

$$\mathbb{E}_{P_0}\left[\left(1 - \frac{R}{\pi_0(X)}\right)^2\middle| X\right] = \frac{1-\pi_0(X)}{\pi_0(X)}.$$

Expanding the variance of $u^\top D_m(O)$ therefore yields

$$\mathrm{Var}\{u^\top D_m(O)\} = \mathrm{Var}\{u^\top D_{\text{eff}}(O)\} + \mathbb{E}_{P_0}\left[\frac{1-\pi_0(X)}{\pi_0(X)}\{u^\top A_0^{-1}\delta_m(X)\}^2\right].$$

The second term is nonnegative and equals zero if and only if $\delta_m(X) = 0$ almost surely. Hence $m_0(X,\theta_0)$ minimizes the asymptotic variance of every linear contrast, so D_{eff} is the efficient influence function within this class. The missing-data efficiency results of Robins et al. (1994, 1995); Robins and Rotnitzky (1995); van der Laan and Robins (2003) identify this same influence function as the semiparametric efficient influence function in the full MAR model. □

Corollary 3 (Two-sample/PPI form)**.** *Suppose the labeled and unlabeled samples arise from the semisupervised model with constant labeling probability ρ_0, and let $S = f(X)$ be a chosen score. Applying theorem 4 with covariate $S = f(X)$ yields the reduced AIPW class*

$$\psi(O;\theta,m,\rho_0) = m(S,\theta) + \frac{R}{\rho_0}\{\varphi(Y,\theta) - m(S,\theta)\}.$$

In the two-sample notation used in the main text, the empirical estimating equation based on the known labeled fraction $\rho_n = n/(n+N)$ can be written as

$$0 = \rho_n \mathbb{P}_n^L\{m(S,\theta)\} + (1-\rho_n)\mathbb{P}_N^U\{m(S,\theta)\} + \mathbb{P}_n^L\{\varphi(Y,\theta) - m(S,\theta)\}.$$

The efficient choice within the reduced score model is

$$m_0(S,\theta_0) = \mathbb{E}[\varphi(Y,\theta_0) \mid S].$$

Proof. Because the semisupervised design has constant labeling probability, R is independent of (X,Y), hence also independent of Y given $S = f(X)$, and

$$\mathbb{P}(R = 1 \mid S) = \rho_0.$$

Applying theorem 4 with covariate $S = f(X)$ and observation probability $\pi_0(S) \equiv \rho_0$ yields the displayed reduced AIPW class and the efficient choice

$$m_0(S,\theta_0) = \mathbb{E}[\varphi(Y,\theta_0) \mid S].$$

At the sample level, the two-sample implementation uses the known design fraction $\rho_n = n/(n+N)$. To rewrite the empirical equation in the two-sample notation, note that

$$\mathbb{P}_M\{m(S,\theta)\} = \rho_n \mathbb{P}_n^L\{m(S,\theta)\} + (1-\rho_n)\mathbb{P}_N^U\{m(S,\theta)\},$$

where $\mathbb{P}_M$ denotes the empirical mean over the pooled sample. Also,

$$\mathbb{P}_M\left[\frac{R}{\rho_n}\{\varphi(Y,\theta) - m(S,\theta)\}\right] = \mathbb{P}_n^L\{\varphi(Y,\theta) - m(S,\theta)\},$$

because $R = 1$ on labeled observations and $R = 0$ on unlabeled observations. Adding the two displays yields exactly

$$0 = \rho_n \mathbb{P}_n^L\{m(S,\theta)\} + (1-\rho_n)\mathbb{P}_N^U\{m(S,\theta)\} + \mathbb{P}_n^L\{\varphi(Y,\theta) - m(S,\theta)\}.$$

□

F Proof of the representation theorem

Proof of theorem 1. Because equation (6) holds for every $h \in \mathcal{F}$, any empirical balancing weight $\widehat{w}(m_n^\star(X)) \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Adding this zero term to the plug-in estimator equation (3) yields equation (7).

For the second claim, since the identity map belongs to $\mathcal{F}$, taking $f(t) = t$ in equation (8) gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\, m_n^\star(X)\big] = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} = \widehat{\psi}_{\text{cal}}.$$

Since $\widehat{w}_n^\star \in \mathcal{F}$, equation (6) also gives

$$\mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0.$$

Therefore,

$$\mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))Y\big] = \mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\, m_n^\star(X)\big] + \mathbb{P}_n^L\big[\widehat{w}_n^\star(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = \widehat{\psi}_{\text{cal}},$$

which is equation (9). □

Lemma 1 (Blockwise residual orthogonality)**.** *Let $B_1, \ldots, B_J$ be the pooled adjacent violator blocks of the isotonic fit, and let $m_{n,\text{iso}}^\star(X_i) = c_j$ for $i \in B_j$. Then*

$$\sum_{i \in B_j} (Y_i - c_j) = 0, \qquad j = 1, \ldots, J.$$

Consequently, for every measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(m_{n,\text{iso}}^\star(X_i))$ is finite for all i,

$$\sum_{i=1}^n h(m_{n,\text{iso}}^\star(X_i))\{Y_i - m_{n,\text{iso}}^\star(X_i)\} = 0.$$

Proof. The fitted values from isotonic regression are constant on pooled adjacent violator blocks. The KKT conditions imply that residuals sum to zero on each block. Since $h(m_{n,\text{iso}}^\star(X_i)) = h(c_j)$ for all $i \in B_j$,

$$\sum_{i=1}^n h(m_{n,\text{iso}}^\star(X_i))\{Y_i - m_{n,\text{iso}}^\star(X_i)\} = \sum_{j=1}^J h(c_j) \sum_{i \in B_j} (Y_i - c_j) = 0.$$

□

Proof of proposition 2. The orthogonality claim equation (10) is exactly the conclusion of Lemma 1. For the risk comparison, note that the identity map belongs to $\mathcal{F}_{\text{iso}}$. Since $\hat{f}$ minimizes the empirical squared loss over $\mathcal{F}_{\text{iso}}$,

$$\mathbb{P}_n^L\big[(Y - m_{n,\text{iso}}^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - \hat{f}_{\text{id}}(m(X)))^2\big] = \mathbb{P}_n^L\big[(Y - m(X))^2\big],$$

where $\hat{f}_{\text{id}}(t) = t$. □

Proof of proposition 3. The normal equations for equation (11) are

$$\mathbb{P}_n^L\{Y - m_{n,\text{lin}}^\star(X)\} = 0 \qquad \text{and} \qquad \mathbb{P}_n^L\big[m(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = 0.$$

Because $m_{n,\text{lin}}^\star(X) = \hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}$, these imply

$$\mathbb{P}_n^L\big[m_{n,\text{lin}}^\star(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m_{n,\text{lin}}^\star(X)$,

$$\mathbb{P}_n^L\big[\hat{w}(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = a\,\mathbb{P}_n^L\{Y - m_{n,\text{lin}}^\star(X)\} + b\,\mathbb{P}_n^L\big[m_{n,\text{lin}}^\star(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = 0.$$

Adding this zero term to equation (12) yields equation (13). □

G Proofs of first-order equivalence between PPI++ and linear calibration

Recall that

$$(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a\, m(X_i) - b\}^2,$$

and define the corresponding linear-calibration estimator by

$$\widehat{\psi}_{\text{lin}} = \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(\widetilde{X}) + \hat{b}_{\text{lin}}\}.$$

Also let $\widehat{\psi}_{++}$ denote the PPI++ estimator obtained by minimizing the empirical influence-function variance over the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, and let $\hat{\lambda}_{++}$ be the resulting selected coefficient. Define

$$\Delta_n := \mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X),$$

and

$$a_0 := \frac{\text{Cov}\{Y, m(X)\}}{\text{Var}\{m(X)\}}.$$

Proof of Proposition 4. The normal equations for the linear regression fit are

$$\mathbb{P}_n^L\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\} = 0$$

and

$$\mathbb{P}_n^L\big[m(X)\{Y - \hat{a}_{\text{lin}} m(X) - \hat{b}_{\text{lin}}\}\big] = 0.$$

The first equation gives

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X).$$

Substituting this into the second equation yields

$$\hat{a}_{\text{lin}} = \frac{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))(Y - \mathbb{P}_n^L Y)\big]}{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big]}.$$

Hence $\hat{a}_{\text{lin}}$ is the ordinary least-squares slope, and under finite second moments,

$$\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2}).$$

Using

$$\hat{b}_{\text{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X),$$

we obtain

$$\begin{aligned}
\widehat{\psi}_{\text{lin}} &= \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(\widetilde{X}) + \hat{b}_{\text{lin}}\} \\
&= \hat{b}_{\text{lin}} + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y - \hat{a}_{\text{lin}} \mathbb{P}_n^L m(X) + \hat{a}_{\text{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\text{lin}}\Delta_n.
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) &= \rho_n \mathbb{P}_n^L\{\hat{\lambda}_{++} m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{\lambda}_{++} m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - \hat{\lambda}_{++} m(X)\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\Delta_n.
\end{aligned}$$

Therefore,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\Delta_n.$$

Under the empirical PPI++ variance criterion over the class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, the selected coefficient satisfies

$$\hat{\lambda}_{++} = \frac{\mathbb{P}_n^L\big[(Y - \mathbb{P}_n^L Y)(m(X) - \mathbb{P}_n^L m(X))\big]}{(1-\rho_n)\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big] + \rho_n \mathbb{P}_N^U\big[(m(\widetilde{X}) - \mathbb{P}_N^U m(\widetilde{X}))^2\big]}.$$

Its population limit is

$$\lambda_0 = \frac{\text{Cov}\{Y, m(X)\}}{\text{Var}\{m(X)\}} = a_0,$$

so standard M-estimation arguments yield

$$\hat{\lambda}_{++} = a_0 + O_p(n^{-1/2}).$$

Combining this with $\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2})$ gives

$$\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = O_p(n^{-1/2}).$$

Next, since $\mathbb{P}_n^L m(X)$ and $\mathbb{P}_N^U m(\widetilde{X})$ are sample means under the same marginal law of X,

$$\Delta_n = O_p(n^{-1/2} + N^{-1/2}).$$

Under the standing assumption $\rho_n = n/(n+N) \to \rho_0 \in (0,1)$, we may write

$$n^{-1/2} = \rho_n^{-1/2}(n+N)^{-1/2}, \qquad N^{-1/2} = (1-\rho_n)^{-1/2}(n+N)^{-1/2},$$

and hence

$$\Delta_n = O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big).$$

Therefore,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} &= (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\Delta_n \\
&= (1-\rho_n)\, O_p(n^{-1/2})\, O_p\Big(\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1/2}\Big) \\
&= O_p\Big((1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\}(n+N)^{-1}\Big).
\end{aligned}$$

Since

$$(1-\rho_n)\rho_n^{-1/2}\Big\{\rho_n^{-1/2} + (1-\rho_n)^{-1/2}\Big\} = \frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}},$$

It follows that

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\Big(\Big\{\frac{1-\rho_n}{\rho_n} + \sqrt{\frac{1-\rho_n}{\rho_n}}\Big\}(n+N)^{-1}\Big).$$

In particular, this implies the simpler bound

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\Big(\frac{1-\rho_n}{\rho_n}(n+N)^{-1}\Big).$$

Therefore, under $\rho_n \to \rho_0 \in (0,1)$,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p\big((n+N)^{-1}\big) = o_p\big((n+N)^{-1/2}\big).$$

This proves the claim. □

Proposition 6 (PPI and AIPW as intercept-only calibration)**.** *Let*

$$\hat{a}_{\text{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^{\star}_{n,\text{const}}(X) := \hat{a}_{\text{const}} + m(X).$$

Then

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Consequently, the PPI estimator

$$\widehat{\psi}_{\text{PPI}} := \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{PPI}} = \mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\},$$

while the standard AIPW estimator

$$\widehat{\psi}_{\text{AIPW}} := \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{AIPW}} = \rho_n \mathbb{P}_n^L\{m^{\star}_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\}.$$

Hence PPI is the unlabeled-only plug-in estimator based on the mean-calibrated score $m^{\star}_{n,\text{const}}$, whereas AIPW is the corresponding pooled plug-in estimator.

Proof. The objective in the definition of $\hat{a}_{\text{const}}$ is quadratic in a, and its first-order condition is

$$0 = \mathbb{P}_n^L\{Y - a - m(X)\}.$$

Therefore

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Using $m^{\star}_{n,\text{const}}(\widetilde{X}) = m(\widetilde{X}) + \hat{a}_{\text{const}}$, we obtain

$$\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} = \widehat{\psi}_{\text{PPI}}.$$

Likewise,

$$\begin{aligned}
\rho_n \mathbb{P}_n^L\{m^{\star}_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^{\star}_{n,\text{const}}(\widetilde{X})\} &= \rho_n \mathbb{P}_n^L\{m(X) + \hat{a}_{\text{const}}\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X}) + \hat{a}_{\text{const}}\} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} \\
&= \widehat{\psi}_{\text{AIPW}}.
\end{aligned}$$

This proves the claim. □

H Proof of Proposition 1

Proof of proposition 1. Let

$$c := \psi_0 - \mathbb{E}_{P_{0,X}}\{f(X)\}.$$

Because c is constant,

$$\widehat{\psi}(f + c) = \widehat{\psi}(f) + \rho_n c + (1-\rho_n)c - c = \widehat{\psi}(f).$$

Therefore $\widehat{\psi}(f) = \widehat{\psi}(\tilde{f})$, where

$$\tilde{f}(X) = f(X) + c = f(X) - \mathbb{E}_{P_{0,X}}\{f(X)\} + \psi_0.$$

By construction,

$$\mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0.$$

Using $\mathbb{E}_{P_0}[Y] = \psi_0$ and $\mathbb{E}_{P_0}\{\tilde{f}(X)\} = \mathbb{E}_{P_{0,X}}\{\tilde{f}(X)\} = \psi_0$, we obtain

$$\begin{aligned}
\widehat{\psi}(f) - \psi_0 &= \widehat{\psi}(\tilde{f}) - \psi_0 \\
&= \rho_n(\mathbb{P}_n^L - P_0)\{\tilde{f}(X)\} + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{\tilde{f}(\widetilde{X})\} + (\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} \\
&= \rho_n(\mathbb{P}_n^L - P_0)\Big\{\tilde{f}(X) - \psi_0 + \rho_n^{-1}\big(Y - \tilde{f}(X)\big)\Big\} \\
&\quad + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\big\{\tilde{f}(\widetilde{X}) - \psi_0\big\}.
\end{aligned}$$

Define

$$D_{f,n}^L(X,Y) := \tilde{f}(X) - \psi_0 + \rho_n^{-1}\{Y - \tilde{f}(X)\}, \qquad D_f^U(\widetilde{X}) := \tilde{f}(\widetilde{X}) - \psi_0.$$

Then

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^n D_{f,n}^L(X_i,Y_i) + \frac{1}{M}\sum_{j=1}^N D_f^U(\widetilde{X}_j).$$

Next,

$$D_{f,n}^L(X,Y) - D_f^L(X,Y) = (\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}.$$

Since $\mathbb{E}_{P_0}\{Y - \tilde{f}(X)\} = 0$, it follows that

$$\begin{aligned}
\frac{1}{M}\sum_{i=1}^n \{D_{f,n}^L(X_i,Y_i) - D_f^L(X_i,Y_i)\} &= \rho_n(\mathbb{P}_n^L - P_0)\Big[(\rho_n^{-1} - \rho_0^{-1})\{Y - \tilde{f}(X)\}\Big] \\
&= \left(1 - \frac{\rho_n}{\rho_0}\right)(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\}.
\end{aligned}$$

Because Y and $f(X)$ have finite second moments, $\tilde{f}(X)$ also has finite second moment, so

$$(\mathbb{P}_n^L - P_0)\{Y - \tilde{f}(X)\} = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Since $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1,

$$\frac{1}{M}\sum_{i=1}^n \{D_{f,n}^L(X_i,Y_i) - D_f^L(X_i,Y_i)\} = o_p(M^{-1/2}).$$

Therefore

$$\widehat{\psi}(f) - \psi_0 = \frac{1}{M}\sum_{i=1}^n D_f^L(X_i,Y_i) + \frac{1}{M}\sum_{j=1}^N D_f^U(\widetilde{X}_j) + o_p(M^{-1/2}),$$

which proves the stated asymptotic linearity. The mean-zero property of D_f^L and D_f^U follows from

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}\{\tilde{f}(X)\} = \psi_0.$$

For the asymptotic variance, write

$$\delta(X) := \tilde{f}(X) - \mu_0(X), \qquad \varepsilon := Y - \mu_0(X).$$

Then

$$\mathbb{E}_{P_0}\{\delta(X)\} = 0, \qquad \mathbb{E}_{P_0}(\varepsilon \mid X) = 0, \qquad \mathbb{E}_{P_0}(\varepsilon^2) = \mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Also,

$$D_f^U(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0 + \delta(\widetilde{X}),$$

and, since $Y - \tilde{f}(X) = \varepsilon - \delta(X)$,

$$\begin{aligned} D_f^L(X,Y) &= \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\} \\ &= \mu_0(X) - \psi_0 + \delta(X) + \rho_0^{-1}\{\varepsilon - \delta(X)\} \\ &= \mu_0(X) - \psi_0 + \rho_0^{-1}\varepsilon + \big(1 - \rho_0^{-1}\big)\delta(X). \end{aligned}$$

Set

$$A(X) := \mu_0(X) - \psi_0, \qquad B(X) := \delta(X), \qquad C(X,Y) := \rho_0^{-1}\varepsilon.$$

Then $\mathbb{E}_{P_0}\{C(X,Y) \mid X\} = 0$, so $C(X,Y)$ is uncorrelated with both $A(X)$ and $B(X)$. Hence

$$\mathrm{Var}\{D_f^L(X,Y)\} = \mathrm{Var}\big\{A(X) + \big(1 - \rho_0^{-1}\big)B(X)\big\} + \mathrm{Var}\{C(X,Y)\},$$

while

$$\mathrm{Var}\{D_f^U(\widetilde{X})\} = \mathrm{Var}\{A(\widetilde{X}) + B(\widetilde{X})\}.$$

Therefore

$$\begin{aligned} &\rho_0\mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} \\ &= \rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) + \rho_0\mathrm{Var}(C). \end{aligned}$$

Expanding the two variances involving A and B gives

$$\begin{aligned} &\rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) \\ &= \{\rho_0 + (1-\rho_0)\}\mathrm{Var}(A) \\ &\quad + \Big\{\rho_0\big(1 - \rho_0^{-1}\big)^2 + (1-\rho_0)\Big\}\,\mathrm{Var}(B) \\ &\quad + 2\left\{\rho_0\big(1 - \rho_0^{-1}\big) + (1-\rho_0)\right\}\mathrm{Cov}(A,B). \end{aligned}$$

Now

$$1 - \rho_0^{-1} = -\frac{1-\rho_0}{\rho_0},$$

so

$$\rho_0\big(1 - \rho_0^{-1}\big)^2 + (1-\rho_0) = \frac{(1-\rho_0)^2}{\rho_0} + (1-\rho_0) = \frac{1-\rho_0}{\rho_0},$$

and

$$\rho_0\big(1 - \rho_0^{-1}\big) + (1-\rho_0) = -(1-\rho_0) + (1-\rho_0) = 0.$$

Thus

$$\rho_0\mathrm{Var}\big\{A + \big(1 - \rho_0^{-1}\big)B\big\} + (1-\rho_0)\mathrm{Var}(A+B) = \mathrm{Var}(A) + \frac{1-\rho_0}{\rho_0}\mathrm{Var}(B).$$

Also,

$$\rho_0\mathrm{Var}(C) = \rho_0 \cdot \rho_0^{-2}\mathbb{E}_{P_0}(\varepsilon^2) = \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big].$$

Since $\mathbb{E}_{P_0}\{B(X)\} = 0$,

$$\mathrm{Var}(B) = \mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

Combining the preceding displays yields

$$\rho_0\mathrm{Var}\{D_f^L(X,Y)\} + (1-\rho_0)\mathrm{Var}\{D_f^U(\widetilde{X})\} = \mathrm{Var}\{\mu_0(X)\} + \rho_0^{-1}\mathbb{E}_{P_0}\big[\mathrm{Var}(Y \mid X)\big] + \frac{1-\rho_0}{\rho_0}\mathbb{E}_{P_0}\big[(\tilde{f}(X) - \mu_0(X))^2\big].$$

This expression is minimized when $\tilde{f}(X) = \mu_0(X)$ almost surely, in particular at $f = \mu_0$. Standard semiparametric characterization results for missing-data models imply that this minimum is the semiparametric efficiency bound in the unrestricted two-sample model; see, for example, Bickel et al. (1993); Robins et al. (1995); Robins and Rotnitzky (1995); van der Laan and Robins (2003); Tsiatis (2006). This completes the proof. □

Lemma 2 (Bounded variation after isotonic post-processing)**.** *Suppose Assumption 3 holds. Let $T := m(X)$ and $S_n := m^\star_{n,\mathrm{iso}}(X)$. Define*

$$\eta(t) := \mathbb{E}_{P_0}[Y \mid T = t], \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s].$$

If $t \mapsto \eta(t)$ has total variation at most V, then, conditional on the labeled sample, the map $s \mapsto \eta_n(s)$ has total variation at most V. Moreover, $|\eta_n(s)| \le C_0$ on the support of S_n.

Proof. We adapt the proof of Lemma 6 in van der Laan et al. (2023). Let $S_n = f_n(T)$, where f_n is the fitted isotonic map. Conditional on the labeled sample, f_n is nondecreasing and piecewise constant. Hence, for each s in the support of S_n, there exists an interval $B_s \subset \mathbb{R}$ such that

$$\{S_n = s\} = \{T \in B_s\}.$$

Therefore

$$\eta_n(s) = \mathbb{E}_{P_0}[Y \mid S_n = s] = \mathbb{E}_{P_0}[\eta(T) \mid T \in B_s].$$

Let $\eta = \eta^+ - \eta^-$ be the Jordan decomposition of η, where η^+ and η^- are nondecreasing and

$$\|\eta\|_{TV} = \|\eta^+\|_{TV} + \|\eta^-\|_{TV} \le V.$$

Then

$$\eta_n(s) = \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] - \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s].$$

If $s_1 < s_2$, then monotonicity of f_n implies every point of B_{s_1} is no larger than every point of B_{s_2}. Hence the conditional distribution of $T \mid T \in B_{s_2}$ first-order stochastically dominates that of $T \mid T \in B_{s_1}$. Therefore, for any nondecreasing function g,

$$\mathbb{E}_{P_0}[g(T) \mid T \in B_{s_1}] \le \mathbb{E}_{P_0}[g(T) \mid T \in B_{s_2}].$$

Applying this with $g = \eta^+$ and $g = \eta^-$, both maps

$$s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s] \qquad \text{and} \qquad s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]$$

are nondecreasing.

Moreover, each of these conditional expectations takes values in the range of the corresponding function, so

$$\big\|s \mapsto \mathbb{E}_{P_0}[\eta^+(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^+\|_{TV}, \qquad \big\|s \mapsto \mathbb{E}_{P_0}[\eta^-(T) \mid T \in B_s]\big\|_{TV} \le \|\eta^-\|_{TV}.$$

Hence

$$\|\eta_n\|_{TV} \le \|\eta^+\|_{TV} + \|\eta^-\|_{TV} = \|\eta\|_{TV} \le V.$$

Finally, if $|\eta(t)| \le C_0$ on the support of T, then

$$|\eta_n(s)| = |\mathbb{E}_{P_0}[\eta(T) \mid T \in B_s]| \le C_0$$

for all s in the support of S_n. □

Lemma 3 (Empirical calibration centers the limit)**.** *Suppose Assumptions 1 and 2 hold and the calibration score equations contain the constant score $h \equiv 1$. Then*

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

Proof. Let $T := m(X)$, and write $m_0 = f_0(T)$, where

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\mathrm{iso}}} \mathbb{E}_{P_0}\big[(Y - f(T))^2\big].$$

Because $\mathcal{F}_{\text{iso}}$ is closed under addition of constants, for every $c \in \mathbb{R}$ the function $f_0 + c$ also belongs to $\mathcal{F}_{\text{iso}}$. Hence the function

$$g(c) := \mathbb{E}_{P_0}\left[(Y - f_0(T) - c)^2\right]$$

is minimized at $c = 0$. Since g is differentiable,

$$0 = g'(0) = -2\,\mathbb{E}_{P_0}[Y - f_0(T)].$$

Therefore

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}[f_0(T)].$$

Recalling that $m_0(X) = f_0(m(X)) = f_0(T)$ and $\psi_0 = \mathbb{E}_{P_0}[Y]$, we conclude

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

□

Proof convention. In the following proofs, fix $\delta > 0$. By $\sup_x |m^\star_{n,\text{iso}}(x)| = O_p(1)$, there exist a deterministic constant $C = C_\delta \geq C_0$ and $n_0 < \infty$ such that the event

$$A_{n,C} := \left\{\sup_x |m^\star_{n,\text{iso}}(x)| \leq C\right\}$$

satisfies $P_0(A_{n,C}) \geq 1 - \delta$ for all $n \geq n_0$. We carry out the deterministic bounded-class arguments on $A_{n,C}$, where all implicit constants may depend on C but not on n. Since δ is arbitrary, the resulting rates and distributional statements hold unconditionally.

Lemma 4 (Isotonic L^2 rate)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$\|m^\star_{n,\text{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof of Lemma 4. Let $T := m(X)$ and let

$$\mathcal{M}_C := \{f(T) : f \in \mathcal{F}_{\text{iso}},\ \|f(T)\|_\infty \leq C\}.$$

By Assumption 2, $g_0(T)$ is bounded by $C_0 \leq C$, and on $A_{n,C}$ we have $m^\star_{n,\text{iso}} \in \mathcal{M}_C$. Since clipping any monotone candidate to $[-C, C]$ preserves monotonicity and can only decrease squared risk against $g_0(T)$, the population isotonic projection m_0 also belongs to $\mathcal{M}_C$. Thus the problem is a one-dimensional isotonic least-squares regression of Y on the score T. Let

$$g_0(T) := \mathbb{E}_{P_0}[Y \mid T].$$

Then, for every $m \in \mathcal{M}_C$,

$$P_0\{(Y - m)^2\} = P_0\{(Y - g_0)^2\} + P_0\{(g_0 - m)^2\},$$

which shows that m_0 is the $L^2(P_T)$-projection of g_0 onto the closed convex set $\mathcal{M}_C$.

Now define the centered class

$$\mathcal{F}_0 := \{g = m - m_0 :\ m \in \mathcal{M}_C\}.$$

For $g \in \mathcal{F}_0$, let $\ell_g(Z) := (Y - m_0(T) - g(T))^2 - (Y - m_0(T))^2$, where $Z = (X, Y)$. Since $m^\star_{n,\text{iso}}$ minimizes the empirical squared risk over $\mathcal{F}_{\text{iso}}$, on $A_{n,C}$ the random element $\hat{g} := m^\star_{n,\text{iso}} - m_0$ belongs to $\mathcal{F}_0$ and satisfies

$$\mathbb{P}^L_n \ell_{\hat{g}} \leq 0 = \mathbb{P}^L_n \ell_0.$$

Moreover,

$$P_0 \ell_g = \|g\|^2_{2,P_{0,X}} - 2P_0\left[(Y - m_0)g\right].$$

Because g is measurable with respect to T, the tower property gives

$$P_0\left[(Y-m_0)g\right] = P_0\left[(g_0-m_0)g\right].$$

Since m_0 is the $L^2(P_T)$-projection of g_0 onto the convex set $\mathcal{M}_C$, the Hilbert-space projection inequality yields

$$P_0\left[(g_0-m_0)(m-m_0)\right] \le 0 \qquad \text{for every } m \in \mathcal{M}_C.$$

Hence

$$P_0\ell_g \ge \|g\|_{2,P_{0,X}}^2,$$

so squared loss has the required quadratic margin around m_0.

We next bound the modulus of continuity of the empirical process indexed by the localized classes

$$\mathcal{F}_\delta := \{g \in \mathcal{F}_0 : \|g\|_{2,P_{0,X}} \le \delta\}.$$

Because every $m \in \mathcal{M}_C$ is bounded by C, every $g \in \mathcal{F}_\delta$ is uniformly bounded, and the standard bracketing bound for one-dimensional monotone classes gives

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta/\varepsilon, \qquad 0 < \varepsilon < \delta.$$

Therefore

$$J_{[]}(\delta, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta^{1/2};$$

see, e.g., van der Vaart and Wellner (1996).

Write $\xi := Y - m_0(T)$, so that $\ell_g = g^2 - 2\xi g$. The class $\{g^2 : g \in \mathcal{F}_\delta\}$ is a bounded Lipschitz image of $\mathcal{F}_\delta$, so by Lemma 3.4.2 of van der Vaart and Wellner (1996) and the same entropy bound,

$$E^\star \sup_{g\in\mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)(g^2)| \lesssim \delta^{1/2}.$$

For the multiplier term, Assumption 2 implies that $Y - g_0(T)$ is sub-Gaussian or subexponential. Since both $g_0(T)$ and $m_0(T)$ are bounded by C, the difference $g_0(T) - m_0(T)$ is bounded, and therefore $\xi = Y - m_0(T)$ is also sub-Gaussian or subexponential. Thus the product class $\xi\mathcal{F}_\delta := \{\xi g : g \in \mathcal{F}_\delta\}$ satisfies the same localized bracketing bound in the Bernstein-type norm used in the subexponential least-squares proof of Bibaut and van der Laan (2019), yielding

$$E^\star \sup_{g\in\mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)(\xi g)| \lesssim \delta^{1/2}.$$

Combining the previous two displays,

$$E^\star \sup_{g\in\mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)\ell_g| \lesssim \delta^{1/2}.$$

We may therefore apply Theorem 3.4.1 of van der Vaart and Wellner (1996) with $d(g) = \|g\|_{2,P_{0,X}}$, quadratic margin $P_0\ell_g \gtrsim d^2(g)$, and modulus $\phi_n(\delta) \asymp \delta^{1/2}$. The fixed-point condition $r_n^2\phi_n(r_n^{-1}) \lesssim \sqrt{n}$ gives $r_n \asymp n^{1/3}$, and hence

$$\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

The preceding display is obtained on $A_{n,C}$. Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same rate holds unconditionally. This is the classical one-dimensional isotonic rate, stated directly in the random-design $L^2(P_{0,X})$ norm relevant for the theorem. □

Theorem 5 (Calibration error of isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\widehat{m}_0(X) := \mathbb{E}_{P_0}[Y \mid m_{n,\mathrm{iso}}^\star(X)].$$

Then

$$\|\widehat{m}_0 - m_{n,\mathrm{iso}}^\star\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Proof. The proof follows that of Theorem 1 in van der Laan et al. (2023); see also van der Laan and Alaa (2025). Let

$$T := m(X), \qquad S_n := m^\star_{n,\mathrm{iso}}(X), \qquad \eta_n(s) := \mathbb{E}_{P_0}[Y \mid S_n = s],$$

and define

$$\Delta_n(X) := \widehat{m}_0(X) - m^\star_{n,\mathrm{iso}}(X) = \eta_n(S_n) - S_n.$$

Conditional on the labeled sample, $h_n(s) := \eta_n(s) - s$ is a measurable function of s. Applying proposition 2 pathwise with $h = h_n$ yields

$$\mathbb{P}^L_n[\Delta_n(X)\{Y - S_n\}] = 0.$$

Also,

$$P_0[\Delta_n(X)\{Y - S_n\}] = P_0[\Delta_n(X)\,\mathbb{E}_{P_0}[Y - S_n \mid S_n]] = P_0(\Delta_n^2) = \|\Delta_n\|^2_{2,P_{0,X}}.$$

Hence

$$\|\Delta_n\|^2_{2,P_{0,X}} = (P_0 - \mathbb{P}^L_n)[\Delta_n(X)\{Y - S_n\}]\,.$$

Fix $\delta > 0$, and work on the event $A_{n,C}$ from the proof convention. On this event, S_n is uniformly bounded by C. By Lemma 2, the function η_n belongs to the bounded-variation class

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$

where $V < \infty$ is the variation bound in Assumption 3. Let

$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\}.$$

Since $S_n = f_n(T)$ for a nondecreasing fitted isotonic map f_n, and composition with a monotone map preserves bounded variation, the function

$$t \mapsto \eta_n\{f_n(t)\}$$

also belongs to $\mathcal{F}_{\mathrm{TV},C}$, after enlarging C if necessary. Thus $\eta_n(S_n) = h_1(T)$ for some $h_1 = f_1 \circ T$ with $f_1 \in \mathcal{F}_{\mathrm{TV},C}$, while $S_n = h_2(T)$ for $h_2 = f_2 \circ T$ with $f_2 = f_n \in \mathcal{F}_{\mathrm{iso},C}$.

Define the localized product class

$$\mathcal{G}_{n,C,\delta} := \Big\{(h_1 - h_2)(Y - h_2) : h_1 = f_1 \circ T,\ h_2 = f_2 \circ T,\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}} \le \delta\Big\}.$$

Then

$$\Delta_n(X)\{Y - S_n\} \in \mathcal{G}_{n,C,\delta_n}, \qquad \delta_n := \|\Delta_n\|_{2,P_{0,X}}.$$

Because bounded-variation functions are differences of bounded monotone functions, standard entropy-preservation results for Lipschitz transformations imply that the localized entropy integral of $\mathcal{G}_{n,C,\delta}$ satisfies

$$J_{[]}(\delta, \mathcal{G}_{n,C,\delta}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Moreover, Assumption 2 implies that $Y - \eta(T)$ is sub-Gaussian or subexponential, and on $A_{n,C}$ the difference $\eta(T) - S_n$ is bounded. Therefore $Y - S_n$ has the same tail type, up to constants. The same localized multiplier empirical-process argument used in the proof of Lemma 4 then gives

$$E^\star \sup_{g \in \mathcal{G}_{n,C,\delta}} |(\mathbb{P}^L_n - P_0)g| \lesssim n^{-1/2}\delta^{1/2}.$$

Set $\varepsilon_n := n^{-1/3}$. For $s \ge 0$, let

$$B_s := \{2^s\varepsilon_n \le \delta_n < 2^{s+1}\varepsilon_n\}.$$

On $B_s \cap A_{n,C}$,

$$2^{2s}\varepsilon_n^2 \le \delta_n^2 \le \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}^L_n - P_0)g|.$$

Therefore, by Markov's inequality,

$$P_0(B_s \cap A_{n,C}) \lesssim \frac{E^\star \sup_{g \in \mathcal{G}_{n,C,2^{s+1}\varepsilon_n}} |(\mathbb{P}_n^L - P_0)g|}{2^{2s}\varepsilon_n^2} \lesssim \frac{n^{-1/2}(2^{s+1}\varepsilon_n)^{1/2}}{2^{2s}\varepsilon_n^2} \lesssim 2^{-3s/2},$$

since $n^{-1/2}\varepsilon_n^{-3/2} = 1$. Summing over $s \geq S$ yields

$$P_0(\delta_n \geq 2^S \varepsilon_n,\ A_{n,C}) \lesssim \sum_{s=S}^{\infty} 2^{-3s/2} \xrightarrow[S\to\infty]{} 0.$$

Because $P_0(A_{n,C}^c) \leq \delta$ for all large n, and $\delta > 0$ is arbitrary, it follows that

$$\|\Delta_n\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

This is exactly the claimed calibration-error rate. □

Proof of Lemma 3. Because $h \equiv 1$ belongs to the calibration score class, equation (6) gives

$$\mathbb{P}_n^L\{Y - m_{n,\mathrm{iso}}^\star(X)\} = 0.$$

Hence

$$\mathbb{P}_n^L Y - \mathbb{P}_n^L m_0(X) = \mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\}.$$

By Cauchy–Schwarz and Lemma 4,

$$P_{0,X}\left|m_{n,\mathrm{iso}}^\star - m_0\right| \leq \|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}),$$

and Lemma 5 gives

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Therefore

$$\mathbb{P}_n^L\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = P_{0,X}\{m_{n,\mathrm{iso}}^\star - m_0\} + (\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = o_p(1).$$

By Assumption 1, the law of large numbers yields $\mathbb{P}_n^L Y \to_p \psi_0$ and $\mathbb{P}_n^L m_0(X) \to_p \mathbb{E}_{P_{0,X}}[m_0(X)]$. Therefore $\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0$. □

Lemma 5 (Centered empirical-process bounds for isotonic calibration)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star(X) - m_0(X)\} = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star(\widetilde{X}) - m_0(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof of Lemma 5. By Lemma 4,

$$\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

To control the centered empirical-process term, work on the event $A_{n,C}$ from the proof convention and consider the localized class

$$\mathcal{F}_{n,C} := \{g = f \circ m - m_0 :\ f \in \mathcal{F}_{\mathrm{iso}},\ \|f \circ m\|_\infty \leq C,\ \|g\|_{2,P_{0,X}} \lesssim n^{-1/3}\}.$$

On $A_{n,C}$, the function $m_{n,\mathrm{iso}}^\star - m_0$ belongs to $\mathcal{F}_{n,C}$. Because m is fixed, this is a localized class of bounded monotone functions of a one-dimensional score. Standard bracketing entropy bounds for monotone classes imply

$$J_{[]}(r, \mathcal{F}_{n,C}, L_2(P_0)) \lesssim r^{1/2}.$$

Applying a local maximal inequality for empirical processes, such as Lemma 3.4.2 of van der Vaart and Wellner (1996), at radius $r_n \asymp n^{-1/3}$ yields

$$\sup_{g \in \mathcal{F}_{n,C}} |(\mathbb{P}_n^L - P_{0,X})g| = O_p\Big(n^{-1/2} r_n^{1/2}\Big) = O_p(n^{-2/3}).$$

In particular,

$$(\mathbb{P}_n^L - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ is arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, $m_{n,\mathrm{iso}}^\star - m_0$ is fixed and the unlabeled sample is independent, so

$$(\mathbb{P}_N^U - P_{0,X})\{m_{n,\mathrm{iso}}^\star - m_0\} = O_p\left(N^{-1/2}\|m_{n,\mathrm{iso}}^\star - m_0\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by Lemma 4. □

Lemma 6 (Centered empirical-process bounds for isotonic post-processing)**.** *Suppose Assumptions 1 to 3 hold. Let*

$$\Delta_n(X) := \widehat{m}_0(X) - m_{n,\mathrm{iso}}^\star(X).$$

Then

$$(\mathbb{P}_n^L - P_{0,X})\Delta_n(X) = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof. By theorem 5,

$$\|\Delta_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Let $V < \infty$ be the variation bound in Assumption 3. Fix $\delta > 0$, and work on the event $A_{n,C}$. On this event, $m_{n,\mathrm{iso}}^\star$ is uniformly bounded by C. By Lemma 2, the function

$$s \mapsto \eta_n(s) := \mathbb{E}_{P_0}[Y \mid m_{n,\mathrm{iso}}^\star(X) = s]$$

has total variation at most V and is bounded by C on the support of $m_{n,\mathrm{iso}}^\star(X)$. Since $m_{n,\mathrm{iso}}^\star(X) = f_n(T)$ with $T = m(X)$ and f_n nondecreasing, it follows that

$$\widehat{m}_0(X) = \eta_n\{f_n(T)\}$$

is a bounded-variation transform of T, while $m_{n,\mathrm{iso}}^\star(X)$ is a bounded monotone transform of T.

Consider the classes

$$\mathcal{F}_{\mathrm{TV},C} := \{f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty \le C,\ \|f\|_{TV} \le V\},$$
$$\mathcal{F}_{\mathrm{iso},C} := \{f : \mathbb{R} \to \mathbb{R} : f \text{ nondecreasing and } \|f\|_\infty \le C\},$$

and define

$$\mathcal{H}_{n,C} := \Big\{h_1 - h_2 : h_1 = f_1 \circ T,\ h_2 = f_2 \circ T,\ f_1 \in \mathcal{F}_{\mathrm{TV},C},\ f_2 \in \mathcal{F}_{\mathrm{iso},C},\ \|h_1 - h_2\|_{2,P_{0,X}}^2 \lesssim n^{-2/3}\Big\}.$$

On $A_{n,C}$, the random function Δ_n belongs to $\mathcal{H}_{n,C}$.

Because bounded-variation functions are differences of bounded monotone functions, the localized bracketing entropy of $\mathcal{H}_{n,C}$ obeys the same bound as in the proof of theorem 5, namely

$$J_{[]}(\delta, \mathcal{H}_{n,C}, L_2(P_0)) \lesssim \delta^{1/2}.$$

Applying the same local maximal inequality at localization radius $n^{-1/3}$ therefore gives, on $A_{n,C}$,

$$|(\mathbb{P}_n^L - P_{0,X})\Delta_n(X)| = O_p(n^{-2/3}).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same bound holds unconditionally.

For the unlabeled sample, conditional on the labeled data, Δ_n is fixed and the unlabeled sample is independent. Hence

$$(\mathbb{P}_N^U - P_{0,X})\Delta_n(\widetilde{X}) = O_p\left(N^{-1/2}\|\Delta_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by theorem 5. □

I Proof of asymptotic linearity

Proof of theorem 2. The proof combines the exact AIPW representation from Theorem 1 with a localized empirical-process bound for the isotonic calibrator class. The latter controls the remainder generated by estimating the monotone post-processing map and yields the $n^{-2/3}$ remainder rate. The overall argument is closely related to the calibrated DML framework of van der Laan et al. (2024c); see also Kennedy (2024) for background on semiparametric influence-function arguments.

By Lemma 3, $P_{0,X} m_0 = \psi_0$. Using theorem 1 with $\widehat{w} \equiv 1$,

$$\widehat{\psi}_{\text{iso}} = \rho_n \mathbb{P}_n^L\{m^\star_{n,\text{iso}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\text{iso}}(\widetilde{X})\} + \rho_n \mathbb{P}_n^L\{Y - m^\star_{n,\text{iso}}(X)\}.$$

Add and subtract the AIPW form built from m_0:

$$\widehat{\psi}_{\text{iso}} = \rho_n \mathbb{P}_n^L\left\{m_0 + \rho_n^{-1}(Y - m_0)\right\} + (1-\rho_n)\mathbb{P}_N^U\{m_0(\widetilde{X})\} + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\text{iso}} - m_0\}.$$

Subtracting ψ_0 gives

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_0 - \psi_0\} \\ + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\text{iso}} - m_0\}.\end{aligned}$$

To replace ρ_n^{-1} by ρ_0^{-1}, note that

$$\rho_n(\mathbb{P}_n^L - P_0)\Big\{(\rho_n^{-1} - \rho_0^{-1})(Y - m_0)\Big\} = \Big(1 - \frac{\rho_n}{\rho_0}\Big)(\mathbb{P}_n^L - P_0)(Y - m_0).$$

The right-hand side is $o_p(M^{-1/2})$ because $\rho_n \to \rho_0$ and

$$(\mathbb{P}_n^L - P_0)(Y - m_0) = O_p(n^{-1/2}) = O_p(M^{-1/2}).$$

Also,

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m^\star_{n,\text{iso}} - m_0\} = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m^\star_{n,\text{iso}} - m_0\} - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\text{iso}} - m_0\}.$$

By Lemma 5, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Thus the remainder $R_{n,N}$, defined as the difference between $\widehat{\psi}_{\text{iso}} - \psi_0$ and the leading empirical-process terms in equation (16), satisfies the bound stated in theorem 2. Since $n^{-2/3} = o(M^{-1/2})$ and $n^{-1/3}N^{-1/2} = o(M^{-1/2})$, this yields equation (16). □

Proof of corollary 1. By theorem 2,

$$\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0) = \frac{1}{\sqrt{M}}\sum_{i=1}^n D^L_{m_0}(X_i, Y_i) + \frac{1}{\sqrt{M}}\sum_{j=1}^N D^U_{m_0}(\widetilde{X}_j) + o_p(1).$$

The labeled and unlabeled samples are independent, each summand has mean zero, and $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1. A standard triangular-array central limit theorem therefore yields the stated asymptotic normal law with variance σ_0^2.

For the variance estimator, let $g_n := m^\star_{n,\text{iso}} - m_0$. By Lemma 4,

$$P_{0,X}(g_n^2) = \|g_n\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Moreover, the proof of Lemma 4 already established the localized empirical-process bound

$$\sup_{g \in \mathcal{F}_\delta} |(\mathbb{P}_n^L - P_{0,X})(g^2)| = O_p(n^{-1/2}\delta^{1/2}),$$

for the squared class $\{g^2 : g \in \mathcal{F}_\delta\}$. Taking $\delta \asymp n^{-1/3}$, which matches the localization radius of g_n, gives

$$(\mathbb{P}_n^L - P_{0,X})(g_n^2) = O_p(n^{-2/3}).$$

Hence

$$\mathbb{P}_n^L(g_n^2) = O_p(n^{-2/3}) = o_p(1).$$

On the boundedness event $A_{n,C}$, conditional on the labeled data, g_n is fixed and $|g_n| \le 2C$, so

$$(\mathbb{P}_N^U - P_{0,X})(g_n^2) = O_p\left(N^{-1/2}\|g_n^2\|_{2,P_{0,X}}\right) = O_p\left(C\,N^{-1/2}\|g_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Therefore

$$\mathbb{P}_N^U(g_n^2) = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big) = o_p(1).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n and arbitrary $\delta > 0$, these bounds hold unconditionally. Also, $\widehat{\psi}_{\text{iso}} \to_p \psi_0$ follows from theorem 2.

Finally,

$$\widehat{D}^L - D_{m_0}^L = (1-\rho_n^{-1})g_n - (\widehat{\psi}_{\text{iso}} - \psi_0) + (\rho_n^{-1} - \rho_0^{-1})(Y - m_0),$$

and

$$\widehat{D}^U - D_{m_0}^U = g_n(\widetilde{X}) - (\widehat{\psi}_{\text{iso}} - \psi_0).$$

Because $\rho_n \to \rho_0$, $Y - m_0$ has finite second moment, and $\mathbb{P}_n^L(g_n^2), \mathbb{P}_N^U(g_n^2) = o_p(1)$, it follows that

$$\mathbb{P}_n^L\Big[\{\widehat{D}^L - D_{m_0}^L\}^2\Big] = o_p(1), \qquad \mathbb{P}_N^U\Big[\{\widehat{D}^U - D_{m_0}^U\}^2\Big] = o_p(1).$$

Therefore $\widehat{\sigma}^2 \to_p \sigma_0^2$. The Wald interval claim then follows by Slutsky's theorem. □

J Proof of post-processing equivalence and reduced-model efficiency

Proof of theorem 3. Let

$$\Delta_n(X) := \widehat{m}_0(X) - m_{n,\text{iso}}^\star(X).$$

Using the definition of the AIPW class together with $\mathbb{P}_n^L\{Y - m_{n,\text{iso}}^\star(X)\} = 0$, we have

$$\begin{aligned}
\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} &= \rho_n\mathbb{P}_n^L\Delta_n(X) + (1-\rho_n)\mathbb{P}_N^U\Delta_n(\widetilde{X}) + \mathbb{P}_n^L\{Y - \widehat{m}_0(X)\} - \mathbb{P}_n^L\{Y - m_{n,\text{iso}}^\star(X)\} \\
&= \rho_n\mathbb{P}_n^L\Delta_n(X) + (1-\rho_n)\mathbb{P}_N^U\Delta_n(\widetilde{X}) - \mathbb{P}_n^L\Delta_n(X) \\
&= (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n.
\end{aligned}$$

Now decompose

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\Delta_n - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\Delta_n.$$

By Lemma 6, the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$. Since $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1, both rates are $o_p(M^{-1/2})$. Therefore

$$\widehat{\psi}_{\text{eff}} - \widehat{\psi}_{\text{iso}} = (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\Delta_n = o_p(M^{-1/2}).$$

This proves the claimed first-order equivalence. In particular, $\widehat{\psi}_{\text{eff}}$ and $\widehat{\psi}_{\text{iso}}$ have the same asymptotic variance. □

Proof of corollary 2. For part (i), if $m_0 = \mu_0$ almost surely, then the asymptotic influence pair in theorem 2 is

$$D^L_{\mu_0}(X,Y) = \mu_0(X) - \psi_0 + \rho_0^{-1}\{Y - \mu_0(X)\}, \qquad D^U_{\mu_0}(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0.$$

By proposition 1, this is the efficient influence pair in the unrestricted two-sample model. Therefore $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

For part (ii), let

$$S := m(X), \qquad \eta_0(s) := \mathbb{E}_{P_0}[Y \mid S = s].$$

proposition 1 applied to the reduced two-sample experiment with scalar covariate S shows that the efficient influence pair in that reduced experiment is

$$D^L_{\eta_0}(S,Y) = \eta_0(S) - \psi_0 + \rho_0^{-1}\{Y - \eta_0(S)\}, \qquad D^U_{\eta_0}(\widetilde{S}) = \eta_0(\widetilde{S}) - \psi_0.$$

If η_0 is monotone increasing, then η_0 belongs to the isotonic class and

$$\mathbb{E}_{P_0}\left[\{Y - g(S)\}^2\right] = \mathbb{E}_{P_0}\left[\{Y - \eta_0(S)\}^2\right] + \mathbb{E}_{P_0}\left[\{\eta_0(S) - g(S)\}^2\right]$$

for every measurable g. Hence the population isotonic regression of Y on the score $S = m(X)$ is exactly $\eta_0(S)$, so

$$m_0(X) = \eta_0\{m(X)\} \qquad \text{almost surely.}$$

Therefore, the influence pair in theorem 2 coincides with the reduced-model efficient influence pair above, so $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the reduced experiment indexed by $m(X)$. □

K Pooled i.i.d. Missing-Data Formulation

The joint two-sample setup can be embedded into a pooled i.i.d. model by introducing a sample-membership indicator $R \in \{0,1\}$, where $R = 1$ denotes labeled observations and $R = 0$ denotes unlabeled observations. In that formulation one observes i.i.d. data $O = (X, R, RY)$ with $\mathbb{P}(R = 1) = \rho_0$, and the calibrated estimator becomes

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_M(m_n^\star) + \mathbb{P}_M\left[\frac{R}{\rho_0}\{Y - m_n^\star\}\right]$$

up to the same calibration identity as in the main text. The corresponding observed-data efficient influence function is

$$m_0(X) - \psi_0 + \frac{R}{\rho_0}\{Y - m_0(X)\},$$

which is the single-sample analogue of the efficient influence pair obtained by applying proposition 1 to the reduced model with scalar covariate $m(X)$. Thus the usual i.i.d. missing-data formulation is a convenient special case, but the main text keeps the two-dataset structure explicit.

L Additional empirical details

This appendix records the benchmark construction used in section 5.2. We use the official `ppi_py` real-data mean-estimation examples and preserve their labeled-sample-size grids. The `forest` experiment uses $n \in \{50, 100, \ldots, 500\}$. The `galaxies` and semisupervised `census_income` benchmarks use $n \in \{50, 155, 261, 366, 472, 577, 683, 788, 894, 1000\}$. For each n, the remaining observations form the unlabeled sample. In the current draft we rerandomize this split 500 times at each n, always evaluating all estimators on the same split and taking the full-sample mean as the benchmark truth.

The estimator set mirrors the original PPI comparison wherever possible. In the paper-facing summaries, we report the labeled-only estimator, the imputation benchmark on binary-outcome tasks, PPI computed using the official `ppi_py` implementation, the classical AIPW estimator based on the same prediction score, clipped

PPI++, and the unclipped efficiency-maximized benchmark AIPW-EM. We then add five calibration-oriented comparators: linear calibration; smooth monotone spline calibration; Platt scaling on the binary-outcome benchmarks; isotonic calibration with a fixed minimum bin size of 10; and an adaptive `AutoCal` selector that chooses among AIPW, linear calibration, monotone spline calibration, and isotonic calibration by cross-validated empirical efficiency. For the binary-outcome benchmarks, we also report Venn–Abers shrinkage. Platt scaling and Venn–Abers are omitted on `census_income` because that outcome is not binary. Including both PPI++ and AIPW-EM lets us separate the underlying efficiency-maximization idea from the specific `ppi_py` clipping rule for the tuning coefficient, while figure 5 isolates the fixed-calibration comparison among PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`.

For the fixed calibration-based plug-in estimators, we fit the calibrator on the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$, transform both the labeled and unlabeled scores, and then compute the pooled plug-in mean together with the same semisupervised influence-style Wald interval used throughout the benchmark code. AIPW uses the same score but averages it over the pooled covariate sample before adding the labeled residual correction, whereas PPI uses only the unlabeled score average. In theory this retains less labeled-sample score information, though in the finite-sample benchmark results the empirical variance gap is often small and not uniform across all datasets. PPI++ and AIPW-EM both apply one-dimensional empirical efficiency maximization to the same score, but PPI++ inherits the official `ppi_py` clipping rule $\lambda \in [0, 1]$ whereas AIPW-EM does not. Linear calibration regresses Y on an intercept and the score, then clips the fitted values to the range of predicted outcomes observed in the labeled data to prevent extrapolation on the unlabeled data. `MonoSpline` instead fits a smooth nondecreasing spline of Y on the score, using monotonicity constraints together with a mild roughness penalty, and then applies the same clipped pooled plug-in construction. Platt scaling fits a logistic regression of the binary outcome on a stabilized logit transform of the score, while the fixed-bin isotonic estimator fits a stepwise monotone calibration map and also clips to the labeled outcome range. The `AutoCal` estimator is a selector rather than a new calibrator: in the experiments it compares AIPW, linear calibration, monotone spline calibration, and fixed-bin isotonic calibration using 20-fold cross-validation and the estimated influence-function variance as the selection criterion. To keep this step inexpensive when the unlabeled sample is very large, the foldwise criterion is evaluated on an unlabeled subsample of size $\min(N, 10n)$. The reported summaries are Monte Carlo averages of bias, empirical variance, MSE, interval coverage, and relative efficiency versus PPI across the repeated random splits. The complete numerical outputs, including dataset-level tables and per-metric plots, are generated automatically by the accompanying reproduction pipeline.

We also retain a supplementary screening sweep for the earlier score-decile gap construction. The screening compares the three public PPE-correctness evaluators (ArmoRM, Athene, Skywork) with the three reproduced tabular PPI benchmarks, always using the same top-versus-bottom-decile construction and the same small-sample calibration workflow. Table 2 helps explain why that decile-gap construction is less attractive for the introduction: the classic tabular benchmarks show much smaller incremental gains, ArmoRM is comparatively mild, and Skywork is too pathological.

Candidate	μ_A	μ_B	score_A	score_B	MSE drop	Note
Athene (PPE correctness)	0.790	0.212	1.751	-1.777	94%	Chosen intro anchor
ArmoRM (PPE correctness)	0.786	0.212	0.052	-0.053	62%	Benign real failure, but too mild for the main toy
Galaxies	0.973	0.011	0.981	0.006	-4%	Real benchmark, but smaller incremental gain
Census income	147977.895	395.886	106255.384	-110.489	8%	Real benchmark, but smaller incremental gain
Forest cover	0.630	0.018	0.727	0.013	4%	Real benchmark, but smaller incremental gain
Skywork (PPE correctness)	0.775	0.229	22.901	-23.555	100%	Very large gain, but driven by pathological raw scaling

Table 2: Supplementary screening sweep for the earlier score-decile gap construction. Each row defines arms A and B as the top and bottom raw-score deciles for that real candidate, reports the resulting human means and raw-score means, and summarizes the held-out squared-error drop from one-dimensional recalibration. The bolded row marks the strongest candidate under that decile-gap screening rule.

Dataset	Smaller labeled-sample regime	Larger labeled-sample regime
`forest`	AIPW and PPI are already hard to beat at the smallest n; among the efficiency-maximized one-dimensional corrections, AIPW-EM stays closer to `LinearCal` than clipped PPI++.	At larger n, all of the score-based methods cluster tightly, with only very small differences among AIPW, `LinearCal`, AIPW-EM, and PPI++.
`galaxies`	AIPW, PPI, and `AutoCal` are very close at the smallest n, while AIPW-EM and `LinearCal` are slightly more competitive than PPI++ in this noisier regime.	At larger n, the leading score-based methods remain tightly clustered, with AIPW-EM typically falling between `LinearCal` and PPI++.
`census_income`	PPI and AIPW are effectively indistinguishable because the unlabeled sample is enormous; AIPW-EM and `LinearCal` already offer modest gains at the smallest n, while PPI++ trails both.	`LinearCal` and AIPW-EM are essentially tied on average and both outperform PPI++ throughout, while PPI and AIPW remain nearly identical.

Table 3: Compact dataset-by-regime summary for the reproduced PPI benchmarks. This table supplements the main-text benchmark paragraph by collecting the finer-grained patterns across labeled-sample regimes.

L.1 LLM evaluator breakdown

The PPE macro-averages in the main text hide substantial evaluator heterogeneity. In particular, the largest raw-proxy failures are driven primarily by Skywork and Athene, while ArmoRM is comparatively benign and leaves all main-text estimators nearly tied. The calibration-based and efficiency-maximized methods are therefore best understood as safeguards against badly scaled reward-model margins, rather than gains that require every evaluator to be severely misaligned.

L.2 LLM ranking supplement

The PPE human-preference track also induces a downstream ranking problem across the selected target models. Using the same repeated labeled/unlabeled splits as in section 5.3, we therefore summarize how well each estimator recovers the target-model ordering induced by the full-data human win shares. For each evaluator, labeled-sample size, replicate, and estimator, we rank the eight selected target models by estimated win share, compare that ordering against the ranking induced by the full-data targets, and then macro-average across evaluators. We report Spearman rank correlation, top-1 identification rate, and top-1 regret, defined as the difference in true win share between the best target model and the model selected by the estimator. The main pattern mirrors the mean-estimation results: calibrated and efficiency-maximized estimators substantially outperform raw PPI/AIPW, while remaining close to labeled-only at small n. At $n = 200$, `AutoCal` attains the highest average Spearman correlation, about 0.905; at $n = 400$, `LinearCal` gives the best overall ranking quality, with Spearman about 0.941 and top-1 identification about 0.646.

M Python Code

Code availability. A public repository containing the `ppi_aipw` package, experiment scripts, and paper assets is available at github.com/Larsvanderlaan/ppi-aipw. Package documentation and worked examples are hosted at larsvanderlaan.github.io/ppi-aipw/.

Below, we provide a minimal, self-contained implementation of the method introduced in this paper.

M.1 Isotonic regression via XGBoost

```
import numpy as np
```

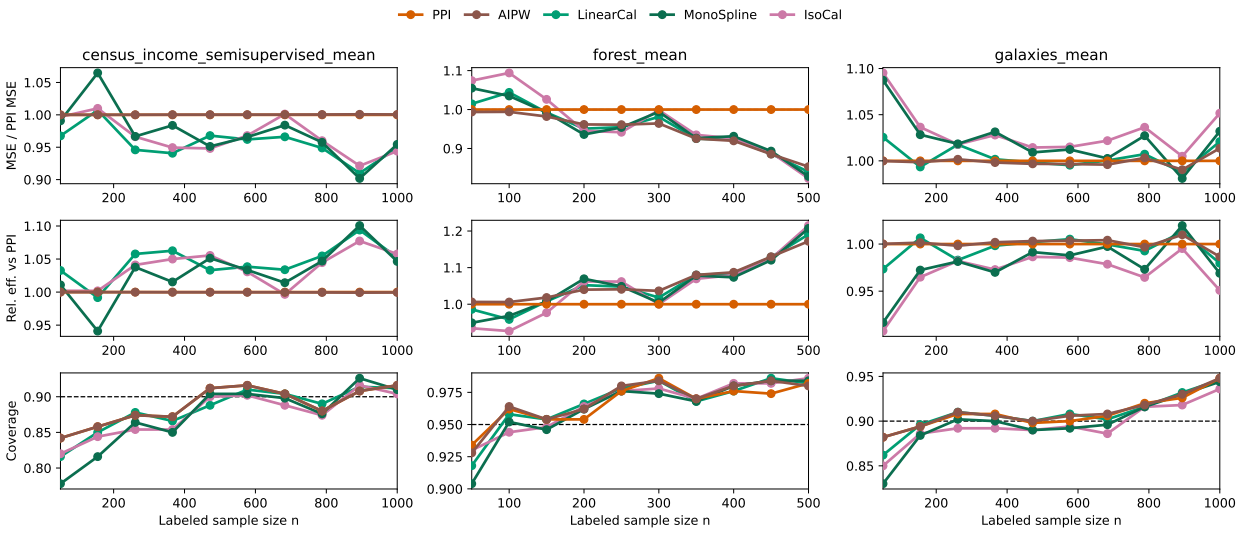

Figure 5: Calibration-focused appendix comparison across the reproduced PPI benchmarks, restricted to PPI, AIPW, `LinearCal`, `MonoSpline`, and `IsoCal`. The panels report normalized MSE relative to PPI, relative efficiency versus PPI, and coverage across labeled-sample-size regimes. This figure isolates the fixed-calibration comparison that is omitted from the main-text benchmark figure for readability.

```
import xgboost as xgb

def isoreg_with_xgboost(x, y, max_depth=15, min_child_weight=20, weights=None):
    """
    Fit a monotone calibrator f so that f(x) is nondecreasing in x.

    Parameters
    ----------
    x : array-like, shape (n,) or (n, 1)
        Predictor used for calibration. In the PPI application this is usually
        the score m(X), so x is typically one-dimensional.
    y : array-like, shape (n,)
        Labeled outcomes.
    max_depth : int, default=15
        Maximum depth for the one-round XGBoost fit.
    min_child_weight : float, default=20
        Minimum total weight in a terminal node.
    weights : array-like, optional
        Optional observation weights.

    Returns
    -------
    predict_fn : callable
        Function mapping new x values to calibrated predictions f(x).
    """
    x = np.asarray(x)
    y = np.asarray(y).reshape(-1)
    x = x.reshape(len(y), -1)

    data = xgb.DMatrix(data=x, label=y, weight=weights)
    params = {
        "max_depth": max_depth,
        "min_child_weight": min_child_weight,
        "monotone_constraints": "(" + ",".join(["1"] * x.shape[1]) + ")",
```

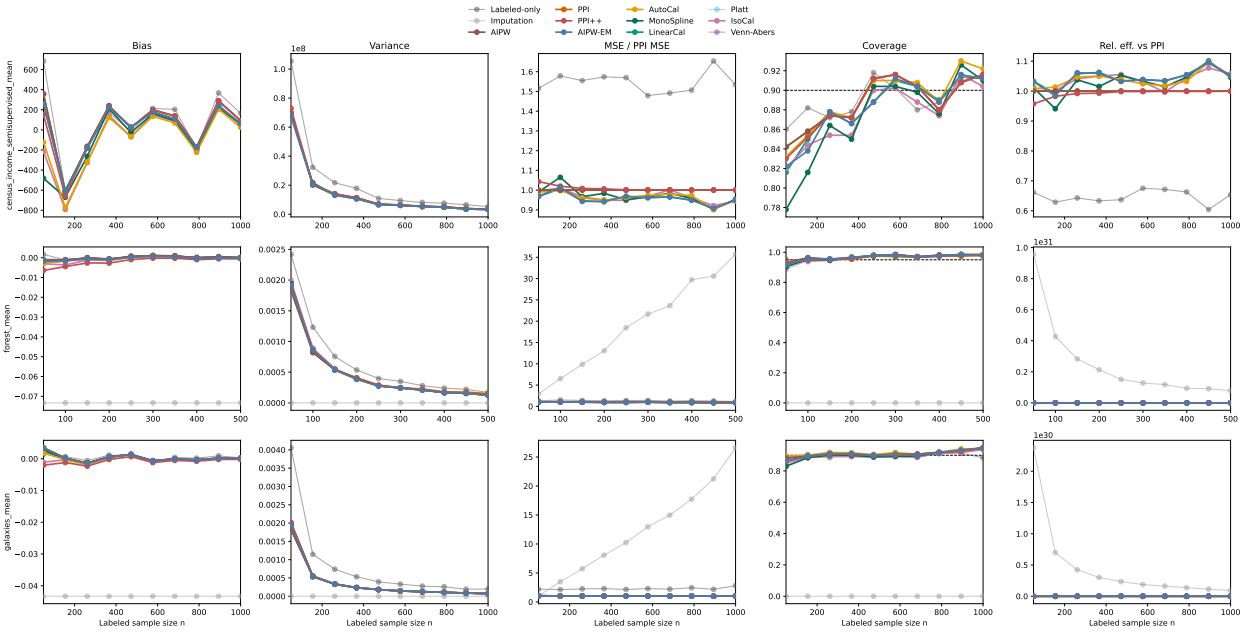

Figure 6: Full diagnostic grid for the reproduced PPI benchmarks, showing bias, empirical variance, normalized MSE relative to PPI, coverage, and relative efficiency versus PPI across labeled-sample-size regimes. In the normalized-MSE and relative-efficiency panels, the dashed horizontal line marks parity with PPI; in the coverage panels it marks the nominal target coverage $1 - \alpha$. `AutoCal`, `MonoSpline`, and `IsoCal` appear alongside PPI, AIPW, PPI++, AIPW-EM, and the fixed calibration rules, while Platt scaling and Venn–Abers appear only on the binary-outcome datasets where they are well defined.

```
        "eta": 1.0,
        "gamma": 0.0,
        "lambda": 0.0,
        "objective": "reg:squarederror",
        "verbosity": 0,
    }
    iso_fit = xgb.train(params=params, dtrain=data, num_boost_round=1)

    def predict_fn(x_new):
        x_new = np.asarray(x_new)
        if x_new.ndim == 1:
            x_new = x_new.reshape(-1, 1)
        data_pred = xgb.DMatrix(data=x_new)
        return iso_fit.predict(data_pred)

    return predict_fn
```

M.2 Isotonic-calibrated plug-in

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
```

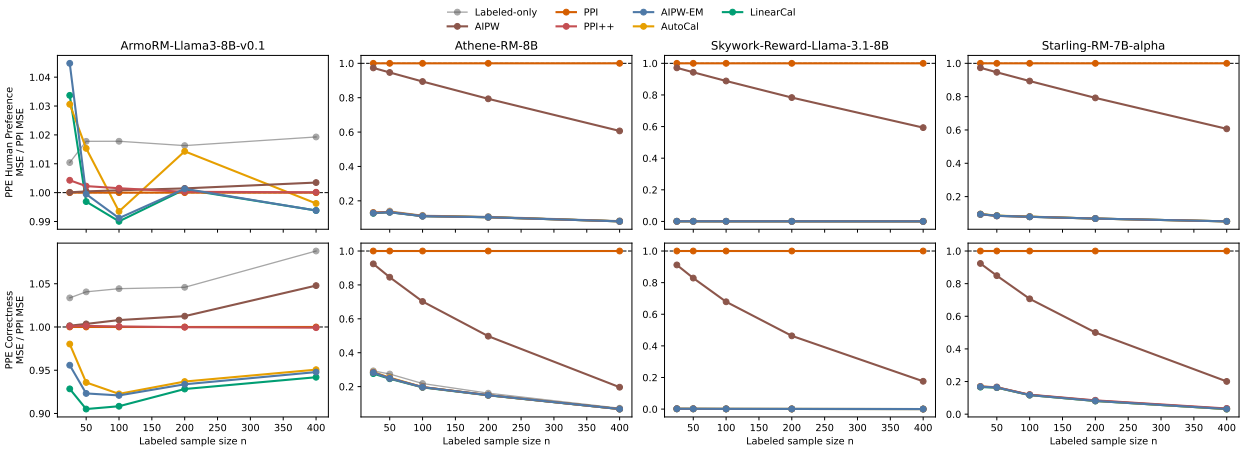

Figure 7: Appendix evaluator-specific PPE summary. Each panel repeats the PPE Human and PPE Correctness comparisons for one public evaluator. The large raw-proxy failures are concentrated in Skywork and Athene, while ArmoRM is much more benign.

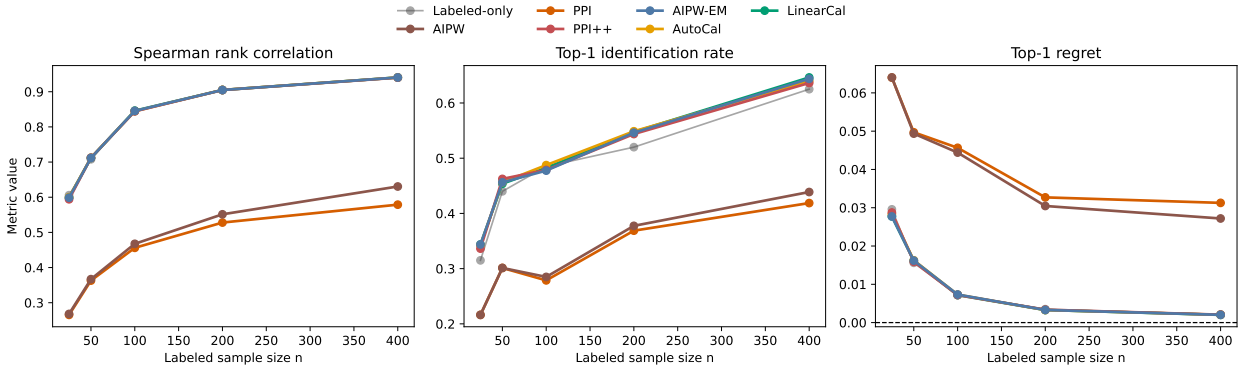

Figure 8: Appendix PPE-ranking supplement from the LLM benchmark, macro-averaged across the public evaluator models. The three panels report Spearman rank correlation, top-1 identification rate, and top-1 regret for recovering the ranking of the eight selected PPE target models from a small labeled sample plus evaluator scores.

```
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)

    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
```

```
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def isotonic_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
    min_obs=20,
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )

    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    calibrator = isoreg_with_xgboost(
        x=score_labeled,
        y=Y_labeled,
        min_child_weight=min_obs,
    )

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
```

```
        "confidence_interval": ci,
    }


# Example:
# fit = isotonic_calibrated_plugin(
#     Y_labeled=Y,
#     X_labeled=X,
#     X_unlabeled=X_tilde,
#     score_fn=m,
# )
```

M.3 Linear calibration

```
import numpy as np
from statistics import NormalDist


def _resolve_scores(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    Y_labeled = np.asarray(Y_labeled).reshape(-1)

    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn together with X_labeled/X_unlabeled, "
            "or provide score_labeled and score_unlabeled."
        )

    score_labeled = np.asarray(score_labeled).reshape(-1)
    score_unlabeled = np.asarray(score_unlabeled).reshape(-1)

    if len(score_labeled) != len(Y_labeled):
        raise ValueError("Y_labeled and score_labeled must have the same length.")

    return Y_labeled, score_labeled, score_unlabeled


def linear_calibrated_plugin(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
```

```
):
    Y_labeled, score_labeled, score_unlabeled = _resolve_scores(
        Y_labeled=Y_labeled,
        X_labeled=X_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )

    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho = n / M

    X_design = np.column_stack([np.ones(n), score_labeled])
    beta_hat, _, _, _ = np.linalg.lstsq(X_design, Y_labeled, rcond=None)

    def calibrator(score):
        score = np.asarray(score).reshape(-1)
        return beta_hat[0] + beta_hat[1] * score

    m_labeled = calibrator(score_labeled)
    m_unlabeled = calibrator(score_unlabeled)

    psi_hat = (n * m_labeled.mean() + N * m_unlabeled.mean()) / M

    if_labeled = m_labeled - psi_hat + (Y_labeled - m_labeled) / rho
    if_unlabeled = m_unlabeled - psi_hat
    var_hat = (
        rho * np.var(if_labeled, ddof=1)
        + (1 - rho) * np.var(if_unlabeled, ddof=1)
    )
    se_hat = np.sqrt(var_hat / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    ci = (psi_hat - z * se_hat, psi_hat + z * se_hat)

    return {
        "estimate": psi_hat,
        "standard_error": se_hat,
        "confidence_interval": ci,
    }


# Example:
# fit = linear_calibrated_plugin(
#     Y_labeled=Y,
#     score_labeled=mX,
#     score_unlabeled=mX_tilde,
# )
```

Dataset	n	N	Estimator	Bias	Variance	MSE	Coverage	RelEff
census_income_semisupervised_mean	50	380041	AIPW	356.4905	69836665.4410	69963750.9366	0.842	1.000
census_income_semisupervised_mean	50	380041	AIPW-EM	260.9833	67728426.6042	67796538.8913	0.822	1.031
census_income_semisupervised_mean	50	380041	AutoCal	-123.2501	69242944.1124	69258134.7078	0.832	1.009
census_income_semisupervised_mean	50	380041	Labeled-only	685.3532	105558402.2427	106028111.3137	0.860	0.662
census_income_semisupervised_mean	50	380041	IsoCal	-215.3832	69697400.2434	69743790.1536	0.820	1.002
census_income_semisupervised_mean	50	380041	LinearCal	260.9294	67622601.4006	67690685.5430	0.816	1.033
census_income_semisupervised_mean	50	380041	MonoSpline	-482.0477	69082916.1270	69315286.1508	0.778	1.011
census_income_semisupervised_mean	50	380041	PPI	356.4473	69834865.9486	69961920.5977	0.842	1.000
census_income_semisupervised_mean	50	380041	PPI++	150.8664	72953053.8328	72975814.4980	0.830	0.957
census_income_semisupervised_mean	577	379514	AIPW	199.8136	6353884.5833	6393810.0648	0.916	1.000
census_income_semisupervised_mean	577	379514	AIPW-EM	177.6067	6114918.2974	6146462.4536	0.912	1.039
census_income_semisupervised_mean	577	379514	AutoCal	134.3746	6201029.6459	6219086.1828	0.910	1.024
census_income_semisupervised_mean	577	379514	Labeled-only	213.2465	9407595.6062	9453069.6946	0.902	0.675
census_income_semisupervised_mean	577	379514	IsoCal	155.0503	6162964.7883	6187005.3977	0.902	1.031
census_income_semisupervised_mean	577	379514	LinearCal	179.8005	6116832.9552	6149161.1594	0.910	1.038
census_income_semisupervised_mean	577	379514	MonoSpline	162.1952	6146069.6864	6172376.9541	0.904	1.033
census_income_semisupervised_mean	577	379514	PPI	199.7932	6351836.6166	6391753.9369	0.916	1.000
census_income_semisupervised_mean	577	379514	PPI++	198.7409	6362047.6345	6401545.5819	0.916	0.998
census_income_semisupervised_mean	1000	379091	AIPW	92.1091	3462051.4638	3470535.5425	0.916	0.999
census_income_semisupervised_mean	1000	379091	AIPW-EM	61.4705	3288235.4312	3292014.0588	0.912	1.052
census_income_semisupervised_mean	1000	379091	AutoCal	25.8207	3289467.9320	3290134.6381	0.922	1.052
census_income_semisupervised_mean	1000	379091	Labeled-only	165.4763	5293296.2822	5320678.6872	0.914	0.654
census_income_semisupervised_mean	1000	379091	IsoCal	51.2697	3271622.4413	3274251.0223	0.904	1.058
census_income_semisupervised_mean	1000	379091	LinearCal	60.7501	3287796.5296	3291487.1058	0.912	1.052
census_income_semisupervised_mean	1000	379091	MonoSpline	60.9626	3306575.6859	3310292.1289	0.910	1.046
census_income_semisupervised_mean	1000	379091	PPI	91.9155	3460071.7259	3468520.1894	0.916	1.000
census_income_semisupervised_mean	1000	379091	PPI++	91.9155	3460071.7259	3468520.1894	0.916	1.000
forest_mean	50	1546	AIPW	-0.0012	0.0018	0.0018	0.928	1.006
forest_mean	50	1546	AIPW-EM	-0.0018	0.0019	0.0019	0.918	0.969
forest_mean	50	1546	AutoCal	-0.0028	0.0019	0.0019	0.924	0.965
forest_mean	50	1546	Labeled-only	0.0017	0.0024	0.0024	0.954	0.762
forest_mean	50	1546	Imputation	-0.0733	0.0000	0.0054	0.000	9564441057429143054280466366464.000
forest_mean	50	1546	IsoCal	-0.0030	0.0020	0.0020	0.930	0.934
forest_mean	50	1546	LinearCal	-0.0016	0.0019	0.0019	0.918	0.986
forest_mean	50	1546	MonoSpline	-0.0021	0.0019	0.0019	0.904	0.949
forest_mean	50	1546	Platt	-0.0003	0.0019	0.0019	0.900	0.948
forest_mean	50	1546	PPI	-0.0013	0.0018	0.0018	0.934	1.000
forest_mean	50	1546	PPI++	-0.0064	0.0019	0.0019	0.928	0.969
forest_mean	50	1546	Venn-Abers	-0.0027	0.0020	0.0020	0.888	0.923
forest_mean	300	1296	AIPW	0.0011	0.0002	0.0002	0.984	1.036
forest_mean	300	1296	AIPW-EM	0.0007	0.0002	0.0002	0.982	1.012
forest_mean	300	1296	AutoCal	0.0006	0.0002	0.0002	0.978	1.002
forest_mean	300	1296	Labeled-only	0.0007	0.0003	0.0004	0.968	0.706
forest_mean	300	1296	Imputation	-0.0733	0.0000	0.0054	0.000	1281441200512508097500322725888.000
forest_mean	300	1296	IsoCal	0.0010	0.0002	0.0002	0.978	1.000
forest_mean	300	1296	LinearCal	0.0010	0.0002	0.0002	0.984	1.018
forest_mean	300	1296	MonoSpline	0.0010	0.0002	0.0002	0.974	1.004
forest_mean	300	1296	Platt	0.0010	0.0002	0.0002	0.982	1.020
forest_mean	300	1296	PPI	0.0011	0.0002	0.0002	0.986	1.000
forest_mean	300	1296	PPI++	-0.0002	0.0002	0.0002	0.980	0.991
forest_mean	300	1296	Venn-Abers	0.0011	0.0002	0.0002	0.978	1.015
forest_mean	500	1096	AIPW	0.0001	0.0001	0.0001	0.980	1.172
forest_mean	500	1096	AIPW-EM	-0.0002	0.0001	0.0001	0.980	1.188
forest_mean	500	1096	AutoCal	-0.0003	0.0001	0.0001	0.986	1.180
forest_mean	500	1096	Labeled-only	0.0001	0.0002	0.0002	0.972	0.879
forest_mean	500	1096	Imputation	-0.0733	0.0000	0.0054	0.000	7819091278107866750853379522 56.000
forest_mean	500	1096	IsoCal	-0.0000	0.0001	0.0001	0.986	1.215
forest_mean	500	1096	LinearCal	0.0001	0.0001	0.0001	0.982	1.191
forest_mean	500	1096	MonoSpline	0.0000	0.0001	0.0001	0.984	1.207
forest_mean	500	1096	Platt	0.0001	0.0001	0.0001	0.984	1.203
forest_mean	500	1096	PPI	0.0000	0.0002	0.0002	0.982	1.000
forest_mean	500	1096	PPI++	-0.0006	0.0001	0.0001	0.978	1.179
forest_mean	500	1096	Venn-Abers	-0.0001	0.0001	0.0001	0.978	1.206
galaxies_mean	50	16693	AIPW	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	AIPW-EM	0.0035	0.0019	0.0020	0.864	0.944
galaxies_mean	50	16693	AutoCal	0.0019	0.0018	0.0018	0.896	0.997
galaxies_mean	50	16693	Labeled-only	0.0014	0.0041	0.0041	0.890	0.451
galaxies_mean	50	16693	Imputation	-0.0433	0.0000	0.0019	0.000	2376289106442477113652488962048.000
galaxies_mean	50	16693	IsoCal	-0.0011	0.0020	0.0020	0.850	0.908
galaxies_mean	50	16693	LinearCal	0.0030	0.0019	0.0019	0.862	0.973
galaxies_mean	50	16693	MonoSpline	0.0026	0.0020	0.0020	0.830	0.917
galaxies_mean	50	16693	Platt	0.0033	0.0019	0.0019	0.856	0.966
galaxies_mean	50	16693	PPI	0.0035	0.0018	0.0018	0.882	1.000
galaxies_mean	50	16693	PPI++	-0.0020	0.0020	0.0020	0.864	0.914
galaxies_mean	50	16693	Venn-Abers	0.0022	0.0020	0.0020	0.830	0.907
galaxies_mean	577	16166	AIPW	-0.0006	0.0001	0.0001	0.906	1.004
galaxies_mean	577	16166	AIPW-EM	-0.0007	0.0001	0.0001	0.906	1.005
galaxies_mean	577	16166	AutoCal	-0.0008	0.0001	0.0001	0.918	0.989
galaxies_mean	577	16166	Labeled-only	-0.0011	0.0003	0.0003	0.910	0.436
galaxies_mean	577	16166	Imputation	-0.0433	0.0000	0.0019	0.000	1873962492481163727822158561 28.000
galaxies_mean	577	16166	IsoCal	-0.0007	0.0001	0.0001	0.894	0.986
galaxies_mean	577	16166	LinearCal	-0.0007	0.0001	0.0001	0.908	1.005
galaxies_mean	577	16166	MonoSpline	-0.0006	0.0001	0.0001	0.892	0.988
galaxies_mean	577	16166	Platt	-0.0007	0.0001	0.0001	0.896	1.002
galaxies_mean	577	16166	PPI	-0.0006	0.0001	0.0001	0.900	1.000
galaxies_mean	577	16166	PPI++	-0.0012	0.0001	0.0001	0.902	0.988
galaxies_mean	577	16166	Venn-Abers	-0.0006	0.0001	0.0001	0.892	0.986
galaxies_mean	1000	15743	AIPW	0.0002	0.0001	0.0001	0.948	0.987
galaxies_mean	1000	15743	AIPW-EM	0.0002	0.0001	0.0001	0.948	0.980
galaxies_mean	1000	15743	AutoCal	0.0001	0.0001	0.0001	0.944	0.977
galaxies_mean	1000	15743	Labeled-only	0.0003	0.0002	0.0002	0.884	0.357
galaxies_mean	1000	15743	Imputation	-0.0433	0.0000	0.0019	0.000	9146652808341511999670190080 0.000
galaxies_mean	1000	15743	IsoCal	0.0001	0.0001	0.0001	0.936	0.951
galaxies_mean	1000	15743	LinearCal	0.0002	0.0001	0.0001	0.946	0.980
galaxies_mean	1000	15743	MonoSpline	0.0002	0.0001	0.0001	0.944	0.969
galaxies_mean	1000	15743	Platt	0.0002	0.0001	0.0001	0.942	0.979
galaxies_mean	1000	15743	PPI	0.0002	0.0001	0.0001	0.948	1.000
galaxies_mean	1000	15743	PPI++	-0.0002	0.0001	0.0001	0.946	0.977
galaxies_mean	1000	15743	Venn-Abers	0.0002	0.0001	0.0001	0.942	0.952

Table 4: Representative numerical summary at the smallest, middle, and largest labeled sample sizes for each reproduced benchmark. The reported quantities are Monte Carlo bias, empirical variance of the point estimator, MSE, Wald-interval coverage, and relative efficiency versus PPI for the displayed benchmark set, including PPI++, AIPW-EM, `AutoCal`, `MonoSpline`, and `IsoCal`.

n	Estimator	Spearman	Top-1	Top-1 regret
25	Labeled-only	0.606	0.315	0.030
25	PPI	0.265	0.216	0.064
25	AIPW	0.268	0.216	0.064
25	PPI++	0.594	0.336	0.029
25	AIPW-EM	0.599	0.344	0.028
25	LinearCal	0.599	0.344	0.028
25	AutoCal	0.600	0.343	0.028
200	Labeled-only	0.905	0.520	0.004
200	PPI	0.528	0.369	0.033
200	AIPW	0.551	0.378	0.030
200	PPI++	0.905	0.544	0.003
200	AIPW-EM	0.905	0.546	0.003
200	LinearCal	0.905	0.548	0.003
200	AutoCal	0.905	0.549	0.003
400	Labeled-only	0.940	0.625	0.002
400	PPI	0.579	0.419	0.031
400	AIPW	0.631	0.439	0.027
400	PPI++	0.940	0.636	0.002
400	AIPW-EM	0.941	0.644	0.002
400	LinearCal	0.941	0.646	0.002
400	AutoCal	0.941	0.640	0.002

Table 5: Appendix PPE-ranking summary at selected labeled sample sizes from the LLM benchmark. Spearman and top-1 are larger-is-better, while top-1 regret is smaller-is-better.